Science, Technology and Society

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MODULE 5
Environmental Awareness

Introduction

The Philippines is no stranger to a variety of environmental problems and hazards due to the combined effects of its topography, people, and global economic status (Punzalan, 2020). As an archipelagic country located west of the Pacific Ocean, the Philippines is surrounded by warm waters which explains the frequency of storms and typhoons. The bad news is, such extreme weather events are only going to get worse due to intensified global warming which is what drives the climate to change. This so-called "climate change" is one of the biggest problems that all humans – regardless of race, gender, and origin – share right now. Experts declare that this problem is something that humans have to deal with collectively.

Learning Outcomes

At the end of this module, students should be able to:

1. Understand what environmentalism means and what it means to be environmentally aware;
2. Learn the differences between environmental science and environmentalism;
3. Understand what climate change is and why Filipinos should be concerned about it; and,
4. Learn the relationships between environmentalism, climate change, and social justice.

Lesson 1. Definition of Environmentalism

Climate change is real (Anderegg et al., 2010). An overwhelming number of publishing climate scientists agree that climate change is already happening, and it is caused by humans (Doran and Zimmerman, 2009). The science behind climate change remains solid (Cook et al., 2016) regardless of our politics and personal beliefs. The fact is whether you believe in it or not, we can all agree that something is wrong with our weather and our climate. Storms and typhoons are becoming stronger and more frequent. Weather events are becoming more extreme and more unpredictable.

What we should all be concerned about is that different individuals and different societies react differently to the changing climate. Not everyone is financially flexible and not everyone has the luxury of choice. Different people adapt in various capacities and some people are most at risk (USAID, 2017). In difficult situations, the poor does not have many options while the rich can adjust until situations go back to normal (Gilmour, 2020).

What is Environmentalism?

Environmentalism is an advocacy on how to preserve, restore, and improve our natural environment including the earth systems (e.g., biodiversity, forests, waters) and earth processes (e.g., climate, weather) (Merriam-Webster, 2010). Environmentalism can also refer to activities that are aimed at protecting the environment from pollution or destruction (Lewis, 2019).

What does it mean to be Environmentally Aware?

Environmental awareness can be characterized as the level of how much we care about environmental issues, which can either be local or global (Aruga, 2020). In simpler terms, environmental awareness can mean just being conscious of the current condition of the environment (Lauwrens, 2020).

What does it mean to be environmentally aware? An environmentally aware person recognizes that her behavior and lifestyle affect the environment in a variety of ways (Anderson, 2019) and to lessen our negative impacts we might need to make beneficial lifestyle changes (Lauwrens, 2020). It has been proposed that environmental awareness is closely associated with

altruism or an unselfish concern for the welfare of other people (Aruga, 2020).

Environmentalism versus Environmental Science

Environmentalism is a type of philosophical doctrine (Lewis, 2019). On the other hand, environmental science is the coming together of different sciences to better understand the environment and solve environmental problems. Environmentalism is more of a social movement while environmental science involves studying of natural environmental processes and how they affect society.

Environmentalism does not require a degree in the environmental sciences and most environmentalists do not have a formal scientific training. A person who believes or engages in the philosophy of environmentalism is called an environmentalist. Environmental scientists, on the other hand, are usually holders of a bachelor's degree in natural sciences such biology, chemistry, botany, geosciences, and environmental science (Browne, n.d.). Environmental scientists do not necessarily become environmentalists and vice versa – but a person can be both. Environmental science can also be a solid foundation for environmentalism.

Importance of Environmental Science

As discussed earlier, being environmentally aware is being able to recognize that our daily lives have varying impacts on the environment (Anderson, 2019) and when taken collectively they can have huge impacts. Being environmentally aware is acknowledging that something is wrong and something must be done. “The first step to solving any problem is recognizing there is one” (The Newsroom Script Episode 1, 2020).

Environmentalists help the public in making more informed decisions about how to use our limited natural resources by doing research, producing reports, writing articles, lectures, issuing press releases, lobbying in congress, fundraising, and campaigns (The Princeton Review, n.d.). Understanding the problem necessitates solutions. If environmental problems can be traced to our lifestyles, then the key to lessening our negative impacts might need to change our lifestyles (Lauwrens, 2020).

Lesson 2. Climate Change

Weather versus Climate

First and foremost: Weather is NOT the same as climate. Weather is what happens today – weather changes every day. Climate is not a single event – climate considers many weather events.

Weather is the temporary condition of the atmosphere, which changes from hour to hour or day to day (National Geographic Society, 2011). On the other hand, climate is an average of weather records over a period of 30 years (Shepherd et al., 2005) which happens in a specific region, as well as its variations and extremes over many years (National Geographic Society, 2011).

Natural Causes of Climate Change

Just like the weather, changes in climate can also be natural. But natural climate change can take hundreds or even thousands of years. For example, the Sahara Desert used to be "green" at some point and several thousand years ago it has experienced rainy weather (National Geographic Society, 2011).

The earth's climate system naturally fluctuates and changes over the course of the Earth's history. Climate change is normal when observed in geologic time scales, e.g., thousands to hundreds of years. We know what influences and changes the climate: 1) volcanic eruptions, 2) ocean currents, 3) the Earth's orbital changes, 4) solar variations and 5) internal variability (NOAA-ESRL, n.d.).

1) Volcanic eruptions cool the climate. Volcanic eruptions cool the climate, but only temporarily. The clouds of dust and ash block out the sun but they fall to the ground eventually, making the cooling effect very short-lived. Although volcanoes release

carbon dioxide (CO2), the average volcanic CO2 are less than 1% of emissions from current human activities.

2) Ocean currents distribute heat around the globe. The waters from the oceans move horizontally and vertically. The ocean currents, also known as circulation systems, are powered by wind, tides, Earth's rotation (Coriolis Effect), energy from the sun, and water density (saltiness). The ocean's circulation system, which circulate the globe in a 1,000-year cycle, distributes heat energy and affects weather and climate.

3) The Earth's orbit is not fixed – it shifts and wobbles. But such changes are so gradual and only happens over thousands of years. Orbital changes affect the beginning and end of ice ages: the last ice age ended about 12,000 years ago while the next cooling cycle may begin in about 30,000 years.

4) Solar energy output also changes. The Sun powers the Earth's climate. But the Sun's energy output can experience some changes over an extended period of time. Changes in the sun's energy output can cause the climate to change. It has been said that a decrease in solar activity triggered the Little Ice Age between 1650 and 1850.

Climate Change Is Happening Now

While the earth's climate system naturally fluctuates and changes over the course of the Earth's history, we have some expectations on the range of temperature (the highs and lows) and how long the temperature cycles occur based on the ocean currents, solar variations, and orbital changes. The current global warming however cannot be explained by natural variations (NOAA-ESRL, n.d.).

It is true that the Earth's climate has changed naturally several times during the planet's history, due to volcanic eruptions and Earth's orbital changes, but the present accelerated increase in temperature and abrupt rise in carbon dioxide levels are unprecedented (IAGLR Fact Sheet, 2009).

Anthropogenic Climate Change

The climate change we are experiencing now is anthropogenic. The term anthropogenic which implies man-made or something that is caused by humans. The massive amounts of greenhouse gases (GHG) released into the atmosphere are the main cause of global warming or what we now know as climate change (Cunningham, 2013).

The presence of GHGs in the atmosphere is normal and they keep the planet warm and livable during nighttime. While GHGs are naturally occurring, the natural concentrations and cycles of GHGs have been disturbed by the growth of industries and domestic markets (Sharma, 2007).

The reason why they are called “greenhouse” is that they kind of act like the glass in a greenhouse – trapping the sun's heat and not allowing it from escaping out. Many of these gases occur naturally, but their concentrations have alarmingly increased in the atmosphere due to human activities (European Union, n.d.). Some of these GHGs are as follows:

1. carbon dioxide (CO2)
2. methane
3. nitrous oxide
4. fluorinated gases

CO2 is the GHG responsible for 64% of man-made global warming and is the most-commonly produced by human activities. Currently, CO2 concentration in the atmosphere is currently 40% higher compared to pre-industrialization levels. Other GHGs trap heat far more effectively than CO2 does but are emitted in smaller quantities. For instance, methane is responsible for 17% of man-made global warming, nitrous oxide for 6% (European Union, n.d.).

Sources of Greenhouse Gases (GHGs)

We know that GHGs trap heat from the incoming solar energy. We have also learned that CO2 is the main GHG that causes the increase in global temperature. We also established that the current CO2 concentration is unusually higher than the pre-industrial levels.

If the culprit behind climate change is GHG, we can trace where they are coming from. If we know where they come from, we can stop them from the source. The following are the causes of the increase CO2 emissions (European Union, n.d.):

1. **Use of fossil fuels.** Fossil fuel use involves burning of coal, oil, and natural gas which produces CO2 and nitrous oxide in the process. The following types of human activities involves burning of fossil fuels:
    - Transportation
    - Heating our homes
    - Industrialization (industries and factories)

2. **Deforestation**. Trees absorb CO2 from the atmosphere and store carbon in their biomass. When cut down, they release their stored carbon in the form of CO2. In some countries like Indonesia, peat swamps and mangroves are cleared and converted into oil palm plantations.
3. **Land use change and land conversion**. Land use change involves conversion of forest uses into urban development, human settlements, and agriculture. It is mostly related to deforestation wherein forests, which are important carbon sinks, are being cleared.
4. **Livestock farming.** Cows and sheep produce large amounts of methane through food digestion.
5. **Agriculture**. Fertilizers contain nitrogen and they produce nitrous oxide emissions.
6. Fluorinated gases produce a very strong warming effect, up to 23,000 times greater than CO2.

Risks of Climate Change (Observations and Predictions)

Carbon dioxide emissions and climate change will negatively affect the ecosystems, hence affecting humans directly or indirectly. According to Sajise et al. (2012), the following are the observed and predicted risks of climate change in the Philippines:

1. **Temperature Increase**. Due to climate change, hot days and warm nights have and will become more frequent while there has been a decrease in cold days and cool nights. This was based on observed values between 1961 to 1990 and 1951 to 2016.

2. **Rainfall Variations (Increase and Decrease).** Rainfall in the Philippines will be highly variable especially on the onset of the rainy season. The overall pattern is decreased rainfall over Luzon and parts of Mindanao are decreasing while there is increased rainfall over the central western part of the country (the Visayan Islands). Some areas in the country have been experiencing more frequent extreme rainfall events while some have decreasing trends in rainfall.

3. **Extreme Weather Events**. There has been an increased frequency of typhoons that enter the Philippine area of responsibility (PAR) between 1990 to 2003. The PAR experiences 20 tropical cyclones on average, most of them coming from the Pacific, with nine (9) making landfalls. Most of these tropical cyclones pass over the Central Visayas region. Over the past 15 years, the recorded extreme weather events have been historical: the strongest typhoon, the most destructive typhoon, the deadliest storm, and the typhoon with the highest recorded 24-hour rainfall.

4. **Sea Level Rise**. Research on sea level rise in major coastal cities have shown a slight upward trend. Ocean waters will rise because warming leads to the thermal expansion of the water in combination with melting glaciers.

   As a country surrounded by water, this is a major concern for the Philippines. Sea level rise could cause the following:

   a) Saltwater intrusion affecting the groundwater
   b) Damage to mangroves
   c) Inundation of coastal farms
   d) Flooding risk to settlement areas

Lesson 3. Climate Change and Social Justice

Climate change threatens human in terms of their physical health and survival. Sources of sustenance may be affected through food and water shortages. This could increase food scarcity and hunger and cause and inequitable access to food. Physical protection may be damaged like loss of home and other properties and, in turn, way of life. The most at risk are thevulnerable sectors like the children, the elderly, poor farmers, and the marginalized communities(Greenpeace International, 2018).

Climate Change Does Not Treat Everyone Equally

Climate change does not have the same effect to all countries around the globe. In fact, the most vulnerable to climate change are the least developed countries and the developing countries (Misra, 2000) like the Philippines. According to a 2013 data, the Philippines was listed in the 2015 Global Climate Risk Index as the number one country that has been affected by climate change (The Climate Reality Project, 2016). Hence, climate change is a real cause for concern for us Filipinos.

Climate change does not treat people equally. When faced with the threats of floods, storm surges, or oncoming typhoons, those who have more money are the ones who can afford to relocate. The rich can easily decide to abandon their belongings, or buy another property, unlike the poor who cannot just leave their prized possessions. Studies also show that the most disproportionately affected are low-income communities, people of color, and women. Worse, people who have been historically marginalized will only be further disadvantaged (Pytel, 2019). A study has shown that high disaster losses lead to high inequality of opportunities (Alisjahbana et al., 2019).

The Poor Suffers the Most

Climate change is an issue that goes beyond politics and economics – it is also one of the biggest human rights issues in human history. As climate change continue to damage the ecosystems, properties, and crops, our access to food, water, shelter, healthcare, and education

are also increasingly becoming more limited (Pytel, 2019). The problem is that not everyone is affected similarly.

We all expect every society in the world to pay a price for global warming. However, the poorest countries and communities will suffer the most from the effects of sea level rise and burning lands including the drastic measures being taken to address the problem (Gilmour, 2020). The urban poor are said to be the most at risk to climate change because they mostly live in temporary shelters that are easily damaged by coastal inundation and storm surges (USAID, 2017).

Climate Change, Human Rights, and Social Justice

Environmental crisis and humanitarian issues are closely related (Gilmour, 2020). According to United Nations, climate change is a human rights issue (UN-OHCHR, 2015). "Human rights refer to the entitlement of all people to be treated equally, to live their life in safety and freedom, and to be protected by their government" (Greenpeace International, 2018).

Climate change is a human rights issue because it affects our well-being – it threatens our very existence as well as the things that sustains our survival: food, water, housing and livelihoods (Amnesty, n.d.). So many of our human rights, such as right to life, health, food, and an adequate standard of living, are adversely affected by climate change. We see the evidence of this, for instance, with each new extreme weather event and the devastation that ensues such as death and the destruction of crops and property.

According to the Climate Reality Project (2016), an extreme weather event costs the Philippines 2% of its gross domestic product (GDP) and another 2% to rebuild infrastructure. Preventing disasters and calamities not only prevent economic damages, but it also prevents poor people from getting poorer (Pytel, 2019).

Climate Change and Environmental Activism

Environmentalism and justice are mutually supportive: when we address global warming, we are also helping the poor, the vulnerable, and most at risk (Wenz, 2007). Therefore, it is

important to identify “hotspots of vulnerabilities”. One way of identifying vulnerabilities is to know the areas where people are living in poverty that are also exposed to climate hazards (Alisjahbana et al., 2019).

When our government officials seem to be uninterested about climate change, people need to act, take a stand, and speak out. Because without further action, climate change will continue to devastate people and the planet, and human rights will continue to be violated. The longer governments wait to take meaningful action, the harder the problem becomes to solve, and the greater the risk that emissions will be reduced through means that increase inequality rather than reduce it (Amnesty, n.d.).

Influential people can also help the promote the fight against climate change. Celebrity endorsements have also helped promote environmentalism. The good thing about using celebrities to advance a cause is that their statements and viewpoints can instantly reach millions. One good example would be the Hollywood actor Leonardo DiCaprio. In 2014, DiCaprio addressed the United Nations to speak about climate change. In the 2016 Oscar Awards, he took the chance to talk about climate change during his acceptance speech. A study has been done about the effect his speech on public interest about the topic. According to statistics, Google searches for climate change soared the day after the awards show (Leas et al., 2016).

Lesson 4. What Climate Change Means for Filipinos

The risks brought about by climate change risks threaten the livelihoods of Filipinos through agriculture, forestry, fisheries, tourism, energy, and mining. Properties and lives are also in danger when weather events become extreme and more unpredictable. The Philippine ecosystems also becomes at risk to extreme weather events such as erosion, landslides, flooding and storm surges. When this happens, many ecosystem services are at risk such as soil stability, shoreline protection, flood control, and biodiversity habitats (Climate Links, 2017).

Why the Philippines is at Risk to Climate Change

According to The Climate Reality Project (2016), there are several factors that makes Philippines even more at risk to climate change. These are the following:

1. **Warmer Waters, Stronger Storms, Sea Level Rise**

The country is surrounded by warm waters. If the temperature increases, more heat leads to more evaporation and more active water cycle. More heat being released into the atmosphere leads to stronger and frequent storms. Warm water also expands and takes up more volume – which is why the sea level rises when the temperature increases.

2. **Lack of Natural Barriers**

The Philippines is composed of 7,000 plus islands and are exposed to the ocean waters. Studies have shown that the mangroves are the most effective barriers against typhoons, protecting the coast from storm surges and erosion. But mangroves are still under threat by logging and conversion to other uses like fishponds.

3. **Disaster Response Needs Improvement**

Based on experience and observations, the Philippines have so much to do to improve its disaster response, such as early warning systems, evacuation plans, and safety shelters before, during, and after disasters and calamities. Even more challenging is the fact that the country's 100 million population are distributed in 7,000 plus islands.

4. **Regularity of Natural Events**

   The Philippines is located along the so-called "typhoon belt" and the Philippine Area of Responsibility is visited by 20 tropical cyclones every year on the average. Ten of these will become typhoons while five will be potentially destructive (de la Cruz, 2016).

Philippine Government's Response

The Philippines have expanded its social protection system after the devastating Typhoon Yolanda in 2013. Such system allowed accessibility of funds to the people who were affected by the disasters, thus preventing them to fall deeper into poverty. This was when the Pantawid Pamilyang Pilipino Program (4Ps Program) was launched, with financial assistance from the World Food Program (WFP) and the UNICEF. The 4Ps enabled the country to provide relief and basic needs of over 105,000 households (Alisjahbana et al., 2019).

The government's 4Ps program has provided urgent assistance to the victims and would later include other aspects of relief and rehabilitation. Given that the country is prone to disasters, the program has been commended for using "the value of social protection at all stages of the disaster management cycle" (Alisjahbana et al., 2019).

Environmental Education Act of 2008 (RA 9512)

To promote environmental awareness, the Philippine Congress passed the Environmental Education Act (Republic Act 9512). The law seeks to promote environmental awareness through environmental education by mandating the integration of environmental education in school curricula at all levels, whether public or private, including in barangay daycare, preschool, non-formal, technical vocational, professional level, indigenous learning and out-of-school youth courses or programs.

Environmental education includes environmental concepts and principles, environmental laws, the state of environment locally and globally, environmental best practices. It also includes the threats of environmental degradation and its impact on human well-being. Most importantly, environmental education highlights the responsibility of the people to the environment and the value of conservation, protection, and rehabilitation of natural resources.

Climate Change Act of 2009 (RA 9729)

In the Philippines, the Climate Change Act (RA 9729) institutionalized the Climate Change Commission (CCC) which was aimed to coordinate government programs and policies related to climate change. The law also developed the National Climate Change Action Plan which provides some guidelines on how to integrate the said Plan into the local development process thereby ensuring that the climate change plans are properly translated into concrete actions at the ground level (World Bank, n.d.).

Disaster Risk Reduction and Management Act of 2010 (RA 10121)

For the longest time, the country's disaster management system has been relief and

response. In 2010, the Philippine Congress legislated the Disaster Risk Reduction and Management (DRRM) Act of 2010 or RA 10121 which transforms our disaster management towards reduction of disaster risk. This transforms the approach from being reactive to being proactive. The idea is that the impacts of disasters can be reduced when the root causes of the risks are addressed: from response to reduction. The law also seeks to strengthen the people's capacity to absorb stress and increase resilience after the disaster event.

The salient features of RA 10121 are as follows (DRRNetPhils, n.d.):

1. It focuses on the most vulnerable sectors (i.e., the poor, the sick, people with disabilities, the elderly, women, and children).
2. It recognizes the important role of local communities.
3. It seeks to strengthen the capacities of local communities.
4. It ensures greater participation from civil society
5. It addresses the root causes of disaster.

Supplemental/Reading Assignment:

1. Search for at least five (5) Filipino environmentalists/environmental activists. Read about what they do/did and what they stand/fought for.

2. Search for at least five (5) foreign environmentalists/environmental. Read about what they do/did and what they stand/fought for.

Assessment Task 5-1

TRUE or FALSE. **Write T if statement is True and write F if the statement is False.**

1. Weather and climate are one and the same.
2. Environmentalism and environmental science are similar.
3. Environmentalists are strictly environmental scientists.
4. Climate change is a human rights issue.
5. Climate change is not real, and it is not yet happening.
6. There is a scientific consensus that the climate change we are experiencing right now is largely anthropogenic or manmade.
7. Environmentalism is an interdisciplinary academic field that integrates physical, biological and information sciences to the study of the environment, and the solution of environmental problems.
8. Environmental science is an advocacy, a belief system, or a philosophical doctrine that the environment is more important than heredity in determining intellectual growth.
9. An environmentally aware person recognizes that her behavior and lifestyle DOES NOT affect the environment.
10. Greenhouse gases are naturally occurring but the current concentrations of CO2 in the atmosphere cannot be linked to natural sources.
11. Trees absorb CO2 from the atmosphere, but they do not store carbon in their biomass. When trees are cut down, their stored carbon is not lost and not released back into the atmosphere.
12. Climate change does not treat people equally because the sector who suffers the most during extreme weather events are the low-income communities, people of color, and women.
13. CO2 is the only greenhouse gas (GHG) responsible for the man-made global warming.
14. The most vulnerable to climate change are the least developed countries (LDCs) andthe developing countries like the Philippines.

Assessment Task 5-2

Choose all the correct answers

1. **Which of the following are sources of greenhouse gases?** Select all that apply.
   a) Burning coal, oil, and gas
   b) Deforestation
   c) Livestock
   d) Agriculture
   e) Sunlight
   f) Transportation and industrialization
   g) Land use change

2. **Which of the following are likely effects/impacts of climate change?** Select all that apply:
   a) Temperature increase
   b) Rainfall variability
   c) Extreme Weather Events
   d) Rise in sea levels
   e) Change in the earth's orbit
   f) Earthquakes
   g) Volcanic eruption
   h) Asteroids
   i) Flooding to coastal settlements
   j) Damage to mangroves
   k) Melting glaciers / melting ice cap

3. **Which of the following are considered vulnerable sectors to climate change?** Vulnerable sectors are the most adversely affected. Select all that apply:
   a) the poor
   b) rich families
   c) world leaders
   d) the elderly
   e) women and children
   f) religious leaders
   g) the sick
   h) politicians
   i) people with disabilities
   j) wealthy pet owners

Assessment Task 5-3

Multiple choice. Select one letter that best corresponds to the answer.

Philippine Laws	**a**	**b**	**c**	**d**
1. Environmental Awareness Law	RA 9152	RA 9729	RA 10121	RA 7929
2. Climate Change Act of 2009	RA 10121	RA 9215	RA 9512	RA 9729
3. Disaster Risk Reduction and Management Act	RA 10121	RA 9279	RA 9512	RA 10211
4. The GHG responsible for 64% of man-made global warming	CO2	methane	nitrous oxides	Fluorinated gases

Summary

The first step to solving any problem is recognizing there is one. Climate change is real and we are already experiencing its impacts. Scientific studies prove that we humans are causing it. As humanity, we share only one planet of which we do not have any substitute for just yet. Hence, we all need to be educated about how our lifestyles are affecting nature.

When we talk about environmental problems like air and water pollution, we also talk about our rights to clean air and safe drinking water being hindered. Additionally, climate change endangers our lives, properties, and livelihoods, generally threatening our rights to a safe living condition. What's more, not everyone has the privilege of financial flexibility and not everyone has the luxury of choice. This is how environmental problems become an issue of human rights.

Since the environment is a shared place for everybody and we are a part of it, it is in our best interests to care about what is happening in our environment – that whatever affects the well-being of others affects our own welfare, too. As students of "science, technology and society", we can apply what we learn in this course to help educate everyone about some of the pressing issues in our environment. Our small actions can make a big difference especially when people think and act the same way, at the same time.

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MODULE 6
Energy Crisis

Introduction

The continued growth of the world's population and the living standards that is dependent on technologies that uses energy is expected to increase the global energy consumption significantly. Energy has become a basic requirement in everyday life and is fueling the global economy (The Global Energy Crisis, 2009). The continued dependence in non-renewable energy sources has great impact on the planet. With the trends of energy crisis, the identified solutions, particularly alternative energy sources, the change in human lifestyle or the ability to adapt to gain energy efficiency, the climate change policy, and the political will in the government's administration, are some of the factors to be considered in addressing the recurring energy crisis.

Learning Outcomes

At the end of the course, the students should be able to:

1. Understand the meaning of energy crisis, the global and local context;
2. Learn the causes of energy crisis;
3. Understand the alternative energy sources and,
4. Learn possible solutions to the energy crisis.

Lesson 1. Definition of Energy Crisis

There is a looming global energy crisis. With several factors at hand that can easily disrupt the supply and demand of energy, it is imperative that we are ready when this event happens. We are slowly understanding that the current energy source can be depleted while the other sources can be harmful to the environment, hence a renewable energy resources that is not harmful to the environment is the best energy alternative that we have.

Details regarding the potential causes and solutions to energy crisis is discussed in this module. Understanding that the energy crisis is related to the depleting natural resources and the increasing human population. A balance in meeting the needs of the people and protection the environment while ensuring a healthy economy is the challenge in tackling the issues related to energy crisis.

Figure 6.1. The first day of gas rationing in California, May 9, 1970

Source: National Archives

Consumers are encouraged by the government to conserve energy and limit unnecessary travel. Car manufacturer started to create smaller vehicles with smaller engines and more fuel-efficient vehicles. These can be seen as positive effects of the 1979 energy crisis (1979 Energy Crisis, 2015).

The 1970s energy crisis is followed by several notable energy related crises in history like the 1990 oil price shock caused by the Gulf War, the 2000-2001 California electric crisis caused by market manipulation and failed deregulation which resulted to multiple large-scale power outages. The 2000 fuel protests in the United Kingdom caused by the increase in the price of crude oil combined with relatively high taxation on road fuel in the UK. This crisis continues in this decade to different counties in the world.

What is Energy Crisis?

Energy crisis is defined as a shortfall in or interruption to the provision of energy supplies according to Collins English Dictionary (2021). It is a severe shortage of one or more energy - producing resources, especially petroleum; (also) a period characterized by such a shortage, in which the price of energy is high and power outages may occur (Oxford Dictionary, 2021).

What is the Global Energy Crisis?

The global energy crisis is the world's demands on the limited natural resources that re used to power industrial society that are diminishing as the demand rises. This is an issue of limited supply since naturally occurring energy sources takes time to replenish and cannot keep up to the continues increase in demand. This crisis is deemed to get worst despite efforts to address the problem. The reason is that there is a limited understanding to the complex causes and solutions to this crisis (Causes, Effects and Solutions to the Growing Problem of Global Energy Crisis, 2014).

The 1970s energy crisis were there was a widespread panic about potential gasoline shortages, and far higher prices for both crude oil and refined products. The supply disruption led to a spike in prices, panic buying, and long lines at gas stations. This happened both in 1973 and 1979.

Oil crisis in the 1970s particularly the oil embargo of 1973-1974 had a deep impact on the everyday life. The oil prices jumped 350% that rippled through the economy of the US. The 1979 energy crisis, on the other hand, occurred in the aftermath of the Iranian Revolution. Iran, a major petroleum exporting country, caused the global supply of crude oil to decline significantly due to the political turmoil in that period. Price per barrel of oil almost doubled within the 12 months of panic buying.

Aside from the Iran issue, the US fiscal policy of regulating oil prices which contributed to higher prices, is to blame for the crisis (1979 Energy Crisis, 2015).

Energy Crisis in the Philippines

The Philippines had a wealth of potential energy sources. In fact, it is the third largest producer of electricity from geothermal sources behind the United States and Indonesia. After many years of placing second in the world ranking, the Philippines is now at third spot after Indonesia (Lectura, 2019). The Philippines has vast undeveloped hydroelectric potential. It has an estimated 1.2 billion tons of coal resources and other sources like solar, animal waste, Agri waste, and other nonconventional sources (Philippines-Energy, 2006).

It was in the 90s when the Philippines was confronted with a crisis caused by insufficient electrical generating capacity. Power interruptions or “Brownouts” as Filipinos refers to power outage or ‘blackouts’ were a regular occurrence up to four hours per day for the Metro manila and the thirty-three provinces in the Luzon power grid (Philippines – Energy (2006). The 620 megawatt nuclear-power plant (Figure 5.1) on the Bataan Peninsula was decided not to be used due to the risk or disaster happening since it is located on a seismic fault. The badly needed expansion of generating capacity in Luzon did not happen and the other facilities like the 110 megawatts geothermal power plant in Albay was delayed due to various reasons like inadequate planning and bureaucratic delays. The series of gas-turbine plants with 500 megawatts capacity was the short-term solution during that time (Philippines-Energy, 2006).

Consumers are encouraged by the government to conserve energy and limit unnecessary travel. Car manufacturer started to create smaller vehicles with smaller engines and more fuel-efficient vehicles. These can be seen as positive effects of the 1979 energy crisis (1979 Energy Crisis, 2015).

The 1970s energy crisis is followed by several notable energy related crises in history like the 1990 oil price shock caused by the Gulf War, the 2000-2001 California electric crisis caused by market manipulation and failed deregulation which resulted to multiple large-scale power outages. The 2000 fuel protests in the United Kingdom caused by the increase in the price of crude oil combined with relatively high taxation on road fuel in the UK. This crisis continues in this decade to different counties in the world.

Figure 6.2. the Bataan Nuclear Power Plant, September 16, 2016

Source: AFP

At present, the Philippines has a looming energy crisis as the Malampaya gas fields are expected to be depleted by 2024 which supplies 30% of Luzon`s energy consumption (Philippine Energy Market, 2021).

Lesson 2. Causes of Energy Crisis

Energy Crises in history are known to have started due to political instability in oil producing countries, wars, market manipulation, and government interventions like tax hikes and regulations of the energy sector. Below are various causes of Energy crisis as listed in the 2014 article entitled Causes, Effects and Solutions to the Growing Problem of Global Energy Crisis.

Overconsumption

Overconsumption of fossil fuels such as oil, gas, and coal has caused strains on our natural resources particularly the depletion of these nonrenewable resources. Pollution of water and air is also associated with overconsumption of these fossil fuels.

Overpopulation

The continued increase in the world population means continued increase in demand for fuels. The more people in the planet, the more energy is consumed. This energy consumption is important to maintain the lifestyle of human beings and the continues flow of the economy at the expense of the health of the environment. Overpopulation means increase of demand to the depleting supply of energy.

Poor Infrastructure

Outdated equipment causes energy shortage that may limit or restrict the production of energy. This aging infrastructure should be upgraded for energy efficiency and to set a high standard of performance of the energy-producing companies.

Unexplored Renewable Energy Options

There are countries whose renewable energy sources are still unused. These countries mostly relied on non-renewable energy sources like coal which adds up to the problem in depleting supply of these nonrenewable energy sources. If utilized, these untapped renewable energy sources, can minimize the dependence on fossil fuels and can also help in reducing

greenhouse gas emissions.

Delay in Commissioning of Power Plants

This delay can be caused by several factors, one of which is the red tape and other political considerations. The delay can affect the expected matching of supply and demand of energy. If the supply doesn't match the demand, it will result in load-shedding and breakdown.

Waste of Energy

Most people do not realize the importance of conserving energy. With a very large global population, the wasted energy coming from inefficient use of electricity by not switching off unused electric fans, lights, or any gadgets, adds up to the pressure on the depleting energy supply.

Poor Distribution System

Frequent tripping and breakdown are a result of a poor distribution system. This causes high energy consumption and a lot of wasted energy.

Major Accidents and Natural Calamities

These major accidents like pipeline burst and natural calamities like the eruption of volcanoes, floods, earthquakes can also cause interruptions to energy supplies and damage energy infrastructures. This will result to a huge gap between supply and demand for energy which leads to increase of prices of essential items, and eventually give rise to inflation.

Wars and Attacks

Energy supply is greatly affected by wars and attacks that happens in the Middle East countries like Saudi Arabia, Iraq, Iran, Kuwait, UAE, or Qatar. An example is the 1990 Gulf war when the price of oil reached its peak causing global shortages and created major problems for energy consumers.

Miscellaneous Factors

Other causes like tax hikes, strikes, military coup, political events, severe hot summers or cold winters can cause a sudden increase in demand for energy and can choke supply. These miscellaneous factors can be any events that significantly affect the balance of energy supply and demand.

Lesson 3. Possible Solutions to Energy Crisis

While we are still finding possible solutions to the energy crisis, mitigation efforts should be seriously considered until we can find the best alternatives to the existing energy sources. Energy conservation is highly encouraging to extend the life span of the depleting energy sources. Below is the list of possible solutions according to the 2014 article entitled Causes, Effects and Solutions to the Growing Problem of Global Energy Crisis.

Move towards Renewable Resources

The solution with the best potential is the shift to renewable energy sources. This move will lessen the dependence of the world's population to the non-renewable energy sources. While we lessen the dependence to these non-renewable energy sources, it is also important that we push for the overall conservation of these resources, or at least what is left of it.

Our current system and infrastructure is designed to consume fossil fuels, it is now the time to start to utilize technology that uses other types of renewable energies – such as steam, solar, and wind. One of the major concerns is the continues use of coal that is harmful to the environment particularly because of the pollution that it brings. Shift to new energy sources is not easy since many of the leading industries use coal as their primary source of power for manufacturing. This is a challenge, but we have to start the shift even in small steps.

Buy Energy-Efficient Products

Buying energy-efficient products seems a small step in solving the energy crisis, but in doing so, with the cooperation of the people, this will have huge impact in reducing energy consumption. The replacement of traditional bulbs with CFLs and LEDs will result to fewer consumption in watts of electricity and the product is guaranteed to last longer. If majority of people across the globe use LEDs and CFLs for residential and commercial purposes, the demand for energy can go down which can help avert energy crisis.

Lighting Controls

This innovative solution of controlling the lighting can also contribute to the reduction of energy consumption. There are a number of new technologies right now that make lighting controls to help save energy in the long run. Preset lighting controls, slide lighting, touch dimmers, integrated lighting controls are few of the lighting controls that can help to conserve energy and reduce overall lighting costs which results to lower energy consumption.

Easier Grid Access

Opening this option to the public can result to increase of potential energy sources. If people can be allowed to plug into the grid and get credit for the power they feed into it, then additional power options can be added to the grid for the benefit of everybody as well as the contributor. The process of getting credit for supplying surplus power back into the grid should be improved so that more people will participate. Subsidy on solar panels should be given to encourage more people to explore renewable options like solar energy.

Energy Simulation

This can be used by big industries or corporation to achieve energy efficiency in their buildings to reduce business energy costs. Energy simulation software are now available and can be used by Engineers, architects, and designers to design the most energy-efficient buildings and to reduce the company's carbon footprint.

Perform Energy Audit

Performing energy audit can help you to reduce your carbon footprint, save energy and money, and avoid energy crisis. This is a process that helps you to identify the areas where your home or office is losing energy and what steps you can take to improve energy efficiency.

Common Stand on Climate Change

This is an issue between the developed and the developing countries. With the regular occurrence of problems related to energy crisis, it is best that all countries should adopt a common stand on climate change. Everybody should focus on reducing greenhouse gas emissions through

an effective cross border mechanism. It is suggested that both developed and developing countries should focus on emissions cuts to halve their emissions from current levels by 2050 to ensure that the Earth will not suffer major energy crisis and negative effects of climate change will still be manageable.

Assessment Task 6-1

True or False. (15 points). Write **T** if statement is True and write **F** if the statement is False.

1. Fossil fuels are renewable energy sources.
2. Coals and nuclear energy sources are better alternative energy sources since it is safe for the environment.
3. Shifting to renewable energy sources like wind and solar is a challenge since most of our facilities are designed to handle fossil fuels.
4. Energy crisis a severe shortage of one or more energy-producing resources, especially petroleum.
5. The 1973 energy crisis is caused by the Iranian Revolution
6. The energy crisis of 1979 led to the development of smaller, more fuel-efficient vehicles.
7. The US fiscal policy of regulating oil prices helped eased the 1979 Energy Crisis which resulted in lower oil prices.
8. At present, the Philippines is ranked as the second largest geothermal energy producer only behind the United States.
9. The Philippines has vast undeveloped hydroelectric potential which can be used as an alternative source by installing solar panels.
10. It was in the 1980s when the Philippines was confronted by the Energy Crisis caused by insufficient electrical generating capacity.
11. The 620-megawatt nuclear-power plant on the Bataan Peninsula was decided not tobe used due to the risk or disaster happening since it is located on a seismic fault.
12. The *Malampaya* gas fields are expected to supply 30% of Luzon`s energy consumption until 2024.
13. The untapped renewable energy sources can minimize the dependence on fossil fuels and can also help in reducing greenhouse gas emissions.
14. Frequent tripping and breakdown are a result of overpopulation and overconsumption.
15. Developed and developing countries should have different stand on Climate Change since they have different energy consumption and have different policies concerning climate change

Assessment Task 6-2

Essay. Please provide concise discussion to the following: (total of 10 points)

1. Please provide potential causes to the potential energy crisis in the Philippines. (5 points)	*Please answer the question using minimum of 100 words, maximum of 200 words*
2. As a citizen, suggest energy conservation strategies (Provide new and innovative solutions). (5 points)	*Please answer the question using minimum of 100 words, maximum of 200 words*

Summary

The Global Energy Crisis of the 1970s is caused by panic and shortage of supply of oil and other petroleum products. This same problem is still true today, we are running out of oil and natural gas. Alternatives like coal and nuclear sources both poses negative environmental impact, both can harm the planet. Renewable energy alternatives like wind and solar are both expensive alternatives.

Energy Crises in history are known to have started due to general and politically related events that causes instability in the supply and demand of energy. To look more closely on the causes of Energy crisis, these ten causes was discussed 1) overconsumption, 2) overpopulation, 3) poor infrastructure, 4) unexplored renewable energy options, 5) delay in commissioning of power plants, 6) waste of energy, 7) poor distribution system, 8) major accidents and natural calamities, 9) wars and attacks, and lastly 10) Miscellaneous factors like tax hikes, strikes, military coup, political events, severe hot summers or cold winters or any events that significantly affect the balance of energy supply and demand.

Having discussed the global and local energy crisis concept and historical overview as well as the various causes that could trigger or contribute to the crisis, the following potential solutions were also discussed. Seven possible solutions were suggested in order to avert the looming global energy crisis associated with the negative effects of climate change; 1) move towards renewable resources, 2) buy energy–efficient products, 3) apply lighting controls, 4) provide easier access to the Grid and provide subsidy to solar power users, 5) conduct energy simulation, 6) perform energy audit, and the last potential solution is 7) have a common stand on climate change for all the countries in the world.

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MODULE 7
The Information Age

Introduction

In the 1970s, Robert Kahn and Vinton Cerf invented TCP/IP, which they integrated into the Department of Defense's ARPANET, and the internet was born. However, it wasn't until the 1990's that it started becoming a widespread phenomenon. In fact, the internet remained largely under government control up until 1984, but when AT&T underwent divestiture in order to get involved with the computer industry, things started to change. While Bill Gates promoted the idea of a personal computer on every desktop, it was the investment of the telecommunication giant that drove the internet forward.

Back then, phone lines could only transmit signals at a snail's pace, and an incoming call would interrupt the connection, heralding the end of a download for many of the nerdy hobbyists of the time. The development of fiber optics and the increased speed of microprocessors meant that signals moved faster, and before long, practically every house on the block was buzzing with that oh so familiar dial-up sound (Sonstrom, 2021).

Learning Outcomes

At the end of this module, students should be able to:

1. Demonstrate understanding of the information age and its impact on our lives
2. Link learned concepts to the development of the information age and its impact on society.
3. Illustrate how the social media and the information age have impacted our lives.
4. Examine human rights in order to uphold such rights in technological ethical dilemmas.

Lesson 1. Historical Development of Information Age (Sonstrom,2021)

Some have begun to call it the Information Revolution. Technological changes brought dramatic new options to Americans living in the 1990s. From the beginning of the decade until the end, new forms of entertainment, commerce, research, work, and communication became commonplace in the United States. The driving force behind much of this change was an innovation popularly known as the Internet.

Personal computers had become widespread by the end of the 1980s. Also available was the ability to connect these computers over local or even national networks. Through a device called a modem, individual users could link their computer to a wealth of information using conventional phone lines. What lay beyond the individual computer was a vast domain of information known as cyberspace.

Lisa

Upon its release in 1983 the Apple "Lisa" computer — named for one of its developers daughters — was supposed to revolutionize personal computing. But interest in "Lisa" was minimal due to its nearly $10,000 price tag and the introduction of the much more affordable "Macintosh" a year later.

The INTERNET was developed during the 1970s by the Department of Defense. In the case of an attack, military advisers suggested the advantage of being able to operate one computer from another terminal. In the early days, the Internet was used mainly by scientists to communicate with other scientists. The Internet remained under government control until 1984.

One early problem faced by Internet users was speed. Phone lines could only transmit information at a limited rate. The development of FIBER-OPTIC cables allowed for billions of bits of information to be received every minute. Companies like INTEL developed faster microprocessors, so personal computers could process the incoming signals at a more rapid rate.

In the early 1990s, the **WORLD WIDE WEB** was developed, in large part, for commercial purposes. Corporations created home pages where they could place text and graphics to sell products. Soon airline tickets, hotel reservations, books, and even cars and homes could be purchased online. Colleges and universities posted research data on the Internet, so students could find valuable information without leaving their dormitories. Companies soon discovered that work could be done at home and submitted online, so a whole new class of telecommuters began to earn a living from home offices unshaven and wearing pajamas.

New forms of communication were introduced. **ELECTRONIC MAIL, or EMAIL**, was a convenient way to send a message to associates or friends. Messages could be sent and received at the convenience of the individual. A letter that took several days to arrive could be read in minutes. Internet service providers like America Online and CompuServe set up electronic chat rooms. These were open areas of cyberspace where interested parties could join in a conversation with perfect strangers.

In 1991, the internet changed from simply a **way to send and receive files,** and became the world wide web of information that we know of today. However, it wasn't until Mosaic was developed by students and researchers at the University of Illinois a year later that people had a user-friendly way to access that web. In 1993, Doom from id Software popularized online gaming, and in 1997, SixDegrees.com (the first website that could be considered a social networking site) launched and electronic companies agreed to make Wi-Fi and wireless internet an industry standard (Sonstrom, 2021).

What's interesting about the **Information Age** is just how quickly technology has developed. The Industrial Age began somewhere around 300 years ago, and lasted right up until about 30 years ago when the internet was in its infancy. While there was a notable acceleration of progress from that of the Agricultural Age, it still took lifetimes before real change in technology took hold. In the early stages of the Information Age, it took decades, and now, technological advancement has accelerated even further, to the point where new technology is released multiple times a year.

With this advanced speed of progress and dearth of easily accessed information, many experts are starting to call for the end of the Information Age. Why? Well, there's a number of different reasons out there. Some experts feel that the amount of information available to the average individual and the rise of Big Data is causing us to stagnate, and it's only a matter of time before something new crops up

Some experts believe we've developed all the technology devoted to information that we possibly can, and that our focus has shifted towards creating better infrastructure instead. With the amount of **research going into self-driving cars, the development of the internet of things, and the push for clean energy**, it's easy to see where this idea has come from. While devices dedicated to information technology are still advancing, at this point, the technology is mostly being refined and improved, and there aren't a lot of new developments (Sonstrom, 2021)..

Maybe Boomers complain about Millennials and their selfies because they're jealous they couldn't **share their experiences as quickly and openly** as we do now?

Others believe we've moved on to an "experience age." They feel that the focus of technological advancements has moved from sharing information to sharing our experiences, and that our technology now revolves around better ways to share those experiences. When you consider the improvements made in cellular phones, the vast strides being made in Virtual Reality technology, and the popularity of social networking, the idea is not without merit. **The way we interact as humans and share our experiences is definitely changing to be a blend of technology and reality**; however, there's a really important concept that this "experience age" theory is missing.

Sharing experiences is still a form of sharing information, and the sharing of information is still a major part of today's society. It's such an important and fundamental ideal that it's been the focus of major scandals, and currently influences America's politics. The fact that informationis so readily available even frightens major nations, like China, who has developed an extensivefirewall that's intended to block negative information about the government from being spread among citizens. Earlier this week, Turkey blocked access to Drop Box, Google Drive, and other major cloud services (Sonstrom, 2021).

Another thing to consider is that the dearth of information available could very well be a driving force behind the development of **artificial intelligence**. Right now, many businesses handle

more data than they know what to do with, the infamous Big Data, which in turn has created the emerging IT position of data scientist.

But the human mind cannot process all the data being generated by businesses all that quickly or efficiently. **Computers, however, can, through machine learning and neural networks**. It's technology that's still getting its feet, and right now its greatest achievement is beating humansat go and Jeopardy, but someday, it may be making big marketing decisions for us (Sonstrom, 2021).

Lesson 2. Applications of Computers in Science and ResearchParticularly During the Corona Virus Pandemic

Applications of Digital Technology in COVID-19 Pandemic Planning and Response

(Whitelaw,Mamas,Topol, & Van Spall, n.d.)

With high transmissibility and no effective vaccine or therapy, **COVID-19 is now a global pandemic.** Government-coordinated efforts across the globe have focused on **containment and mitigation, with varying degrees of success**.

Countries that have maintained low COVID-19 per-capita mortality rates appear to share strategies that include early surveillance, testing, contact tracing, and strict quarantine. The scale of coordination and data management required for effective implementation of these strategies has—in most successful countries—relied on adopting digital technology and integrating it into policy and health care (Whitelaw, Topol, & Van Spall, n.d.)

COVID-19, an infectious disease caused by the novel severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), is a global pandemic. With high transmissibility, a case fatality rate greater than 1%, and no effective antiviral therapy or vaccine, the mainstay of pandemic management has been containment and mitigation.1 However, despite relying on established public health principles, countries across the world have had varying degrees of success in managing the burden of COVID-19.

Digital health technology can facilitate pandemic strategy and response in ways that are difficult to achieve manually (figure).2 Countries such as South Korea have integrated digital technology

into government-coordinated containment and mitigation processes—including surveillance, testing, contact tracing, and strict quarantine—which could be associated with the early flattening of their incidence curves.3 Although South Korea has incurred only 0·5 COVID-19deaths per 100 000 people,3, 4 the USA, with three times as many intensive care unit beds per 100 000 people and ranked number one in pandemic preparedness before the COVID-19 pandemic, has sustained ten times as many deaths per capita.

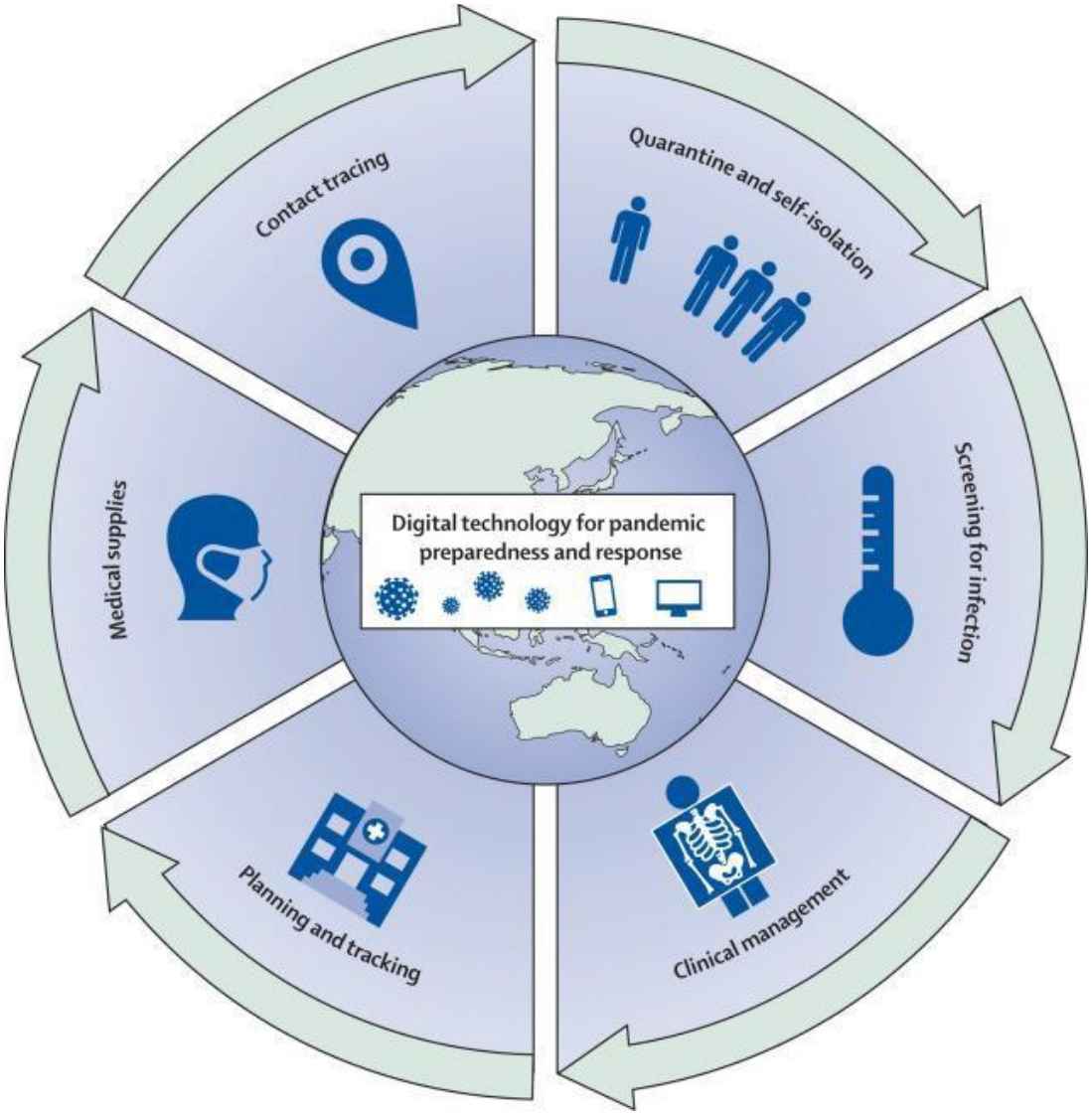

Figure 7.1 Digital Technology as a Tool for Pandemic Preparedness and Response

Source: (Whitelaw et al, n.d.)

Table 7.1 Digital Technology Initiatives Used In Pandemic Preparedness And Response

	Functions	Digital technology	Countries	Advantages	Disadvantages
Tracking	Tracks disease activity in real time	Data dashboards; migration maps; machine learning; real-time data from smartphones and wearable technology	China; Singapore; Sweden; Taiwan; USA	Allows visual depiction of spread; directs border restrictions; guides resource allocation; informs forecasts	Could breach privacy; involves high costs; requires management and regulation
Sc reening for infection	Screens individuals and populations for disease	Artificial intelligence; digital thermometers; mobile phone applications; thermal cameras; web-based toolkits	China; Iceland; Singapore; Taiwan	Provides information on disease prevalence and pathology; identifies individuals for testing, contact tracing, and isolation	Could breach privacy; fails to detect asymptomatic individuals if based on self-reported symptoms or monitoring of vital signs; involves high costs; requires management and regulation; requires validation of screening tools
Contact tracing	Identifies and tracks individuals who might have come into contact with an infected person	Global positioning systems; mobile phone applications; real-time monitoring of mobile devices; wearable technology	Germany; Singapore; South Korea	Identifies exposed individuals for testing and quarantine; tracks viral spread	Could breach privacy; might detect individuals who have not been exposed but have had contact; could fail to detect individuals who are exposed if the application is deactivated, the mobile device is absent, or Wi-Fi or cell connectivity is inadequate
Quarantine and self-isolation	Identifies and tracks infected individuals, and implements quarantine	Artificial intelligence; cameras and digital recorders; global positioning systems; mobile phone applications; quick response codes	Australia; China; Iceland; South Korea; Taiwan	Isolates infections; restricts travel	Violates civil liberties; could restrict access to food and essential services; fails to detect individuals who leave quarantine without devices
Clinical management	Diagnoses infected individuals; monitors clinical status; predicts clinical outcomes; provides capacity for telemedicine services and virtual care	Artificial intelligence for diagnostics; machine learning; virtual care or telemedicine platforms	Australia; Canada; China; Ireland; USA	Assists with clinical decision-making, diagnostics, and risk prediction; enables efficient service delivery; facilitates patient-centred, remote care; facilitates infection control	Could breach privacy; fails to accurately diagnose patients; involves high costs; equipment may malfunction

Source: (Whitelaw, et.al, n.d.)

Planning and tracking (Whitelaw et al., n.d.)

Big data and artificial intelligence (AI) have helped facilitate COVID-19 preparedness and the tracking of people, and so the spread of infection, in several countries. Tools such as migration maps, which use mobile phones, mobile payment applications, and social media to collect real-time data on the location of people, allowed Chinese authorities to track the movement of people who had visited the Wuhan market, the pandemic's epicenter. With these data, machine learning models were developed to forecast the regional transmission dynamics of SARS-CoV-2 and guide border checks and surveillance.

As soon as China reported the outbreak, Taiwan initiated health checks for airline travelers from Wuhan, integrating data from immigration records with its centralized, real-time national health insurance database.8 This integration allowed health-care facilities to access patients' travel histories and identify individuals for SARS-CoV-2 testing and tracking.8 Taiwan's proximity to Wuhan, China, made the region particularly susceptible to COVID-19, but its efficient use of big data is credited for the low number of cases and deaths.

Swedish Health Services, a USA-based health-care organization, has developed a platform for health-care workers to report real-time data on volumes of patients with COVID-19, personal protective equipment, staffing, ventilator usage, and other resource information.9 This information has been shared across its hospitals to track the status of facilities, allocate health-care resources, and increase hospital bed capacity.

The need to track COVID-19 has fueled the innovation of data dashboards that visually display disease burden. UpCode uses data provided by the Singapore Ministry of Health to depict infection trends across age, sex, and location, and to plot the recovery time of infected individuals.10 The Johns Hopkins University (MD, USA) coronavirus dashboard and the web-based platform HealthMap provide up-to-date visuals of COVID-19 cases and deaths around the globe.4 AI algorithms allow the effect of the climate to be incorporated into the projections.11

AI is not without limitations and requires training with COVID-19 datasets. Most of the AI predictive models so far have used Chinese samples, which might not be generalizable. In addition to the absence of historical training data, social media and other online traffic have created noise in big-data sets, potentially producing over fitted or so-called lucky good fit models.1

this noise must be filtered before accurate trends and predictions can be discerned. The accuracy, validity, and reliability of each AI forecast should be assessed when interpreting projections.

Screening for Infection (Whitelaw et al., n.d.)

China uses free, web-based and cloud-based tools to screen and direct individuals to appropriate resources. High-performance infrared thermal cameras set up in Taiwanese airportsare used to capture thermal images of people in real time, rapidly detecting individuals with a fever.8 In Singapore, people have their temperature measured at the entries of workplaces, schools, and public transport. The data from the thermometers is tracked and used to identify emerging hot spots and clusters of infection where testing could be initiated.

Unlike most other countries, Iceland has launched widespread testing of asymptomatic individuals. Using mobile technology, Iceland collects data on patient-reported symptoms and combines these data with other datasets such as clinical and genomic sequencing data to reveal information about the pathology and spread of the virus. This approach has added to the knowledge base regarding the prevalence and transmission of asymptomatic COVID-19. To date, Iceland has had the highest per-capita testing rate and among the lowest per-capita COVID-19 mortality rate. Other countries offering widespread testing include Germany and South Korea.

In the USA, a private company has used digital thermometers to collect real-time data on clusters of febrile illness, and a national study is capturing resting heart rate with a smartwatch application, which could be able to identify COVID-19 emerging outbreaks.20 these initiatives are either enterprise-driven or investigational and are not integrated into policy and practice.

Systematic screening technologies are expensive and require trained personnel, restricting their uptake in many countries. The incubation period and the relatively high prevalence of asymptomatic infection compared with other infectious diseases limits the effectiveness of digital systems that screen vital signs or self-reporting of symptoms. Researchers at the European Centre for Disease Prevention and Control estimate that a majority of passengers from Chinese cities would not be detected by screening because of these factors.

Contact Tracing

South Korea has implemented tools for aggressive contact tracing, using security camera footage, facial recognition technology, bank card records, and global positioning system (GPS) data from vehicles and mobile phones to provide real-time data and detailed timelines of people's travel. South Koreans receive emergency text alerts about new COVID-19 cases in their region, and people who could have been in contact with infected individuals are instructed to report to testing centers and self-isolate. By identifying and isolating infections early, South Korea has maintained among the lowest per-capita mortality rates in the world.

Singapore has launched a mobile phone application that **exchanges short-distance Bluetooth signals when individuals are in proximity to each other.** The application records these encounters and stores them in their respective mobile phones for 21 days. If an individual is diagnosed with COVID-19, Singapore's Ministry of Health accesses the data to identify contacts of the infected person.10 Like South Korea, Singapore has maintained one of the lowest per-capita COVID-19 mortality rates in the world.

Germany has launched a **smartwatch application that collects pulse, temperature, and sleep pattern data to screen for signs of viral illness**. Data from the application are presented on an online, interactive map in which authorities can assess the likelihood of COVID-19 incidence across the nation. With widespread testing and digital health interventions, Germany has maintained a low per-capita mortality rate, relative to other countries, despite a high prevalence of cases.

Contact tracing applications are not without pitfalls not all exposure requires quarantine, such as when the exposed individuals are wearing personal protective equipment or are separated by thin walls penetrable by mobile phone signals. On the other hand, relevant exposure could be missed when individuals do not carry their mobile phones or are without mobile service.In addition, researchers at Oxford University (UK) have suggested that 60% of a country's population would need to use a contact tracing application for it to be an effective mitigation strategy.

Quarantine and self-isolation

The indiscriminate lockdowns for infection control in several countries have had severe socioeconomic consequences. With digital technology, quarantine can be implemented in individuals who have been exposed to or infected with the virus, with less strict restrictions imposed on other citizens. China's quick response (QR) code system, in which individuals are required to fill out a symptom survey and record their temperature, allows authorities to monitor health and control movement. The QR code serves as a COVID-19 health status certificate and travel pass, with color codes representing low, medium, and high risk; individuals with green codes are permitted to travel unrestricted, whereas individuals with red codes are required to self-isolate for 14 days. China also uses AI-powered surveillance cameras, drone-borne cameras, andportable digital recorders to monitor and restrict the gathering of people in public.

In Australia, international travelers were quarantined in hotels on arrival, with travelers from Wuhan quarantined off the Australian mainland. In new legislation, individuals breaching quarantine will be forced to wear tracking devices, with fines levied for further instances of breaking the restrictions. In Taiwan, electronic monitoring of home-quarantined individuals is facilitated through government-issued mobile phones tracked by GPS; in the event of a breach in quarantine, this so-called digital fence triggers messages to the individual and levies fines. In South Korea, individuals in self-isolation are instructed to download a mobile phone application that alerts authorities if they leave their place of isolation. In Hong Kong, people in self-isolation are required to wear a wristband linked through cloud technology to a database that alerts authorities if quarantine is breached. Iceland has launched a mobile phone solution to monitor individuals with COVID-19 and ensure that they remain in self-isolation.

Mobile phone solutions for quarantine enforcement can be bypassed if individuals leave their quarantine location without their devices. Self-reported surveys such as those used in QR code systems only work when individuals are symptomatic and report their symptoms accurately. However, such technological innovations could provide benefits when used in combination with other strategies.

Clinical Management

AI can facilitate rapid diagnosis and risk prediction of COVID-19. **A cloud-based AI-assisted CT service is used to detect COVID-19 pneumonia cases in China**. This technology processes CT images in seconds, differentiating COVID-19 from other lung diseases and speeding up the diagnostic process substantially. COVID-Net, an open-source deep convolutional neural network design available to clinicians across the globe, can quickly detect COVID-19 cases from other lung diseases on chest x-rays. Machine learning algorithms developed in China can predict the likelihood of developing acute respiratory distress syndrome and critical illness among infected patients. These prediction models can guide clinical decision-making and resource allocation, identifying regions and hospitals in need of critical care resources and medical supplies.

Virtual care platforms, using video conferencing and digital monitoring, have been us ed worldwide to deliver remote health care to patients as a means of reducing their exposure to SARS-CoV-2 in health-care institutions. In Canada, clinician-to-patient video visits have increased from approximately 1000 visits per day in February, 2020, to 14 000 per day by mid- May. Countries such as the USA and Australia have also harnessed digital technology to provide remote care to patients with chronic conditions or with mild or moderate COVID-19 illness in theirhomes. If implemented and delivered appropriately, virtual care can increase health-care accessduring the pandemic and after, but possible risks could include misdiagnoses, equipment malfunction, privacy breaches, and costs to the health-care system.

Risks of Digital Technology

Digital health initiatives can amplify socioeconomic inequalities and contribute to health-care disparities. **Digital technology typically involves the use of the internet and mobile phones.**Although 4 billion people used the internet worldwide in 2019, usage was disproportionally higherin high-income areas than in low-income and middle-income areas (82% in Europe vs 28% in Africa). Even within high-income countries, susceptible groups, such as those in low-income neighborhoods or remote regions, might not have access to broadband signals, smartphones, or wearable technology such as smartwatches. To effectively implement digital technology globally, interventions should be tailored to the target regions; broadband access requires federal and private sector investment in technology and infrastructure. At a regional level, subsidized mobile

phone plans, loaner devices, free Wi-Fi hotspots, and training programmers could provide

temporary solutions to these disparities. In regions without infrastructure or sufficient funds to support cellular and data coverage, automated applications and devices that do not require continuous network access should be considered.

Several digital health interventions, particularly those that track individuals and enforce quarantine, can infringe on privacy, while increasing risk among individuals with mental illness or restricted access to food or water. **Government-implemented surveillance and control can instill fear and threaten civil liberties. To balance the need for contact tracing and privacy**, European authorities have proposed that data be retained for only 14 days, the period of possible viral transmission, and that non-essential digital measures be lifted once the pandemic ends. Some European countries are deploying an opt-in smartphone tracking application with anonymized data, no central database, and no GPS information. The appropriate concerns about privacy and data security are potentially offset by facilitating a return to normal routine without a rebound in infections.

The Covid-19 Pandemic as Research Driving Force (Michopaulou, 2020)

The COVID-19 health crisis has temporarily affected people's lives dramatically by changing jobs, education, research and entertainment since the beginning of 2020. However, some temporary solutions like teleworking and distance learning have led to more permanent changes which are likely to continue once the pandemic ends. The future that emerges from the health crisis is not a scene from a science fiction movie, but it is happening now! For example, it's visible how the needs of using information and telecommunication technology as well as the concern of risking our health by using public transportation have fostered research in a few months.

Research culture is changing

The rapid sharing of the coronavirus genetic sequence worldwide enabled real-time progress in the understanding of the new COVID-19 disease and fostered an international cross-border mobilization of scientists in the public and private sector research to immediately start working on its identification.

What is clear from recent months is the value of a shared goal to bring people together

especially within the research community. The COVID-19 pandemic highlighted the crucial need for international cross-border scientific collaboration to develop diagnostics, vaccines and treatments to tackle health emergencies, as well as the need for maintaining and supporting this collaboration and open access and exchange of samples, data and information on an ongoing basis, and not only during emergency situations.

John Ioannidis, Professor of Medicine and of Epidemiology and Population Health at Stanford University says that "before the COVID-19 pandemic, it had been painstakingly slow to make progress in **research openness and reproducible research practices. The pandemic may help overcome some of this inertia by making an all too obvious case that unless we share maximally our data, methods, and tools, and unless we are fully transparent, we run the risk ofbeing overwhelmed by tenuous, spurious or outright wrong, if not even fraudulent, information**. In this regard, the retraction of major papers from top journals such as the Lancet and New EnglandJournal of Medicine, was an eye-opener. Much needs to be done, but we need to grasp this unique opportunity to transform our research practices and overall investigative culture."

Since the start of the outbreak, scientists have embraced new ways of working together at an unprecedented rate to answer the most pressing questions about COVID-19. It is characteristic that researchers and publishers pulled together to make the coronavirus -related research immediately accessible. In March 2020 more than 30 publishers committed to make COVID-19 content freely available and reusable.

Constantinos Demetzos, Professor of Pharmaceutical Nanotechnology of the National & Kapodistrian University of Athens says that "it could not be done differently, given the crucial situation and the effects on society and the economy", noting the fact that within a few months the research on the vaccine made leaps and bounds and created know-how, which will allow scientists in the future to treat other diseases. What the research revealed during the pandemic period? "**What becomes obvious from the latest updates on the development of vaccines is theeffectiveness of combining the science of molecular biology with nanotechnology. In the case of SARS-CoV-2 vaccine, nanotechnology offers the Nano technological platform** (i.e. Lipid Nano Particles, LNP's) which proves effective in carrying while protecting the bioactive molecule (i.e. mRNA). The pandemic spurred scientists from different disciplines to work together and to eventually bring important results such as the vaccine against SARS-CoV-2.

In the current tragic pandemic context, the velocity of scientific networks appeared to be

unprecedented and this led to important achievements for the scientific community. We come to witness the creation of big scientific databases and knowledge repositories due to the vast production of publications and the accumulation of knowledge coming both from clinical trials and from laboratory results. This knowledge will prove useful for future health research".

New Opportunities, Non-Visible Before the Pandemic Crisis

Historical experience shows that a **global crisis of this magnitude creates new opportunities for science and economics**, leading – especially in the case of science – to roads that were not visible before. Dr. Alexandros Papaderos, Deputy Head Office for Research & Innovation, Head of Innovation, Technical University of Munich (TUM) claims that: "In the corona crisis, which is **characterized by uncertainties about the economic and social consequences, science is receiving a particularly high level of attention. Universities and research institutes have to use this chance to shape global developments such as technological leaps, new pandemics, other disasters and the foreseeable climate changes, socially, ecologically and economically sustainable.** We should use this opportunity to be in a better position on many issues than before the crisis. The years before the crisis and especially the first few months of the crisis, have shown that start-ups and spin-outs are making important contributions to society being solution providers and innovation drivers. With a new speed and willingness to experiment, many innovative solutions were introduced into society and industry and positive experiences were made. The state and the established companies should use the know-how, the innovative services and the products of these young companies much more intensively in the future. In particular, when the economy becomes stronger and more sustainable after the corona crisis, then a more intensive paradigm shift will be necessary, not thinking and acting in silos of the different types of companies, but to strengthen the cooperation between different parts of the economy. The need for this has been shown at the latest by the current pandemic".

"The **biggest lesson that pandemic has taught us is the need to be prepared**", says the Vice President-elect of the European Research Council (ERC), Nektarios Tavernarakis, Chairman of the Board of Directors at the Foundation for Research and Technology-Hellas and Professor of Molecular Systems Biology at the Medical School of the University of Crete, in Heraklion, Greece, talking to Greek press. "Take for example the RNA vaccines, which were recently developed and

shown to be more than 90% effective. The technology used is based on more than 20 years of prior research. We wouldn't have the capability to develop a vaccine within a few months without this technology. This exceptional feat of vaccine production within a few months is attributed to the long-term investment in frontier research", claims Professor Tavernarakis adding that: "ERC focuses on supporting cutting-edge, blue skies research, precisely because it believes that without it, modern human societies are essentially defenseless and vulnerable against unpredictable threats. Recent events show clearly that we cannot afford, and should not be 'unarmed' in the face of a natural disaster or a new pathogen."

Acceleration of Digital Services

The pandemic **accelerated the creation in a short time of new digital services that facilitate everyday life but also create better conditions in work, economy and education**. For George K. Karagiannidis, a Professor of Digital Telecommunication Systems at the Electrical & Computer Engineering Department of Aristotle University of Thessaloniki, Greece, a leading scientist selected from Clarivate Analytics as 2020 Web-of-Science Highly Cited Researcher (for the 6th consecutive time) this is a new reality: "As an example, this reality is changing the way we have designed telecommunications networks so far, as capacity requirements are now being transferred from the business and school-university to our home, and this change refers to fixed as well as to wireless networks". Also, he mentioned "**the research during the pandemic period raised the need for an interdisciplinary approach and collaboration between the scientific fields of Medicine, Biology and Informatics.** Without this cooperation it would not be possible in this very short period of time to understand the behavior of the covid-19, to closely monitor the pandemic and to take measures, but importantly to create vaccines."

His research team named "Wireless Communications and Information Processing (WCIP) Group" is recognized as one of the leading research teams in Europe in the field of Telecommunications and Signal Processing, and is active in two research areas, which are expected to have significant social impact as well. The **first concerns technology trends towards the 5G and after 5G wireless networks, where his research concerns the development of new services and the improvements of the energy and data-rate performance of devices, such as smartphones, tablets and sensors .** As an example, his team is one of the first in the world workingon the utilization of **indoor lighting as a wireless network, like WiFi**. These systems are called LiFi and we are going to see them work soon, maybe in 2021.

The second research area of his team concerns **signal processing for biomedical applications.** Here we will mention its contribution to the development of the next generation of Cochlear Implants for people with very serious hearing problems. In June 2020 he published the idea of the first exclusively optical (all optical) cochlear implant. The great advantages of this implant compared to the existing ones are the better quality of hearing, the minimization of the risk of damage to the auditory nerve and the lower energy consumption, which allows a longer life of the implant.

Professor George K. Karagiannidis adds that the technological changes that have taken place over the last 20 years, those during the pandemic crisis included, are unprecedented in human history. We live in this historical time, we are part of these events and for this reason we cannot understand the frightening magnitude of the changes that are taking place: "I am optimistic that the 4th industrial revolution will be for the human's benefit. But at the same time, I am concerned about the great challenges and questions that arise. For example, I recently read a study in which it is estimated that 85% of jobs for 2030 have not yet been invented.

So, how can a theory of economics and employment be developed today if we do not know the working model that will exist in 10 years? "And Professor Karagiannidis adds: "today theoretical tools in philosophy, politics and economics to answer these challenges and questions do not exist, because any such theory cannot be grounded if it does not take into account the dramatic and rapid changes that are taking place in the field of technology, and especially in informatics and telecommunications. That is, today there is more than ever a need for interdisciplinary development of new theoretical tools, even between seemingly unrelated scientific fields, such as computer science, political science, economics and philosophy, between which until now there was no – or there is minimal – communication."

Lesson 3. Impact of Living in the Information Age during theCorona Virus Pandemic

The Social Impact of the Pandemic (United Nations Department of Economic and Social Affairs,2020)

The UN Department of Economic and Social Affairs (UN DESA) is a pioneer of sustainable development and the home of the Sustainable Development Goals (SDGs), where each goal finds its space and where all stakeholders can do their part to leave no one behind.

UN DESA through the Division for Inclusive Social Development (DISD), monitors national and global socio-economic trends, identifies emerging issues, and assesses their implications for social policy at the national and international levels. To this end, we are a leading analytical voice for promoting social inclusion, reducing inequalities and eradicating poverty. According to the United Nations Department of Economic and Social Affairs, the world is facing a global health crisis unlike any in the 75-year history of the United Nations — one that is killing people, spreading human suffering, and upending people's lives. But this is much more than a health crisis. It is a human, economic and social crisis. The coronavirus disease (COVID-19), which has been characterized as a pandemic by the World Health Organization (WHO), is attacking societies at their core.

The COVID-19 outbreak affects all segments of the population and is particularly detrimental to members of those social groups in the most vulnerable situations, continues to affect populations, including people living in poverty situations, older persons, persons with disabilities, youth, and indigenous peoples.

Early evidence indicates that that the health and economic impacts of the virus are being borne disproportionately by poor people. For example, homeless people, because they may be unable to safely shelter in place, are highly exposed to the danger of the virus. People without access to running water, refugees, migrants, or displaced persons also stand to suffer disproportionately both from the pandemic and its aftermath – whether due to limited movement, fewer employment opportunities, increased xenophobia etc.

If not properly addressed through policy the social crisis created by the COVID-19 pandemic may also increase inequality, exclusion, discrimination and global unemployment in the medium and long term. Comprehensive, universal social protection systems, when in place, play a much durable role in protecting workers and in reducing the prevalence of poverty, since they act as automatic stabilizers. That is, they provide basic income security at all times, thereby enhancing people's capacity to manage and overcome shocks.

As emphasized by the United Nations Secretary-General, during the launch of a COVID-19 Global Humanitarian Response Plan on 23 March 2020 "We must come to the aid of the ultra-vulnerable – millions upon millions of people who are least able to protect themselves. This is a matter of basic human solidarity. It is also crucial for combating the virus. This is the moment to step up for the vulnerable."

Mental Health during COVID-19: Signs that you and Your Child May Need More Support(American Academy of Pediatrics, 2020)

1. Changes in mood that are not usual for your child, such as ongoing irritability, feelings of hopelessness or rage, and frequent conflicts with friends and family.
2. Changes in behavior, such as stepping back from personal relationships. If your ordinarily outgoing teen shows little interest in texting or video chatting with their friends, for example, this might be cause for concern.
3. A loss of interest in activities previously enjoyed. Did your music-loving child suddenly stop wanting to practice guitar, for example? Did your aspiring chef lose all interest in cooking and baking?
4. A hard time falling or staying asleep, or starting to sleep all the time.
5. Changes in appetite, weight or eating patterns, such as never being hungry or eating all the time.
6. Problems with memory, thinking, or concentration.
7. Less interest in schoolwork and drop in academic effort.

8. Changes in appearance, such as lack of basic personal hygiene (within reason, since many are doing slightly less grooming during this time at home).
9. An increase in risky or reckless behaviors, such as using drugs or alcohol.
10. Thoughts about death or suicide, or talking about it (see "a word about suicide risk," below).

Challenges and Opportunities during the Outbreak (American Academy of Pediatrics, 2020)

Social distancing to slow the spread of COVID-19 can be especially hard for teens, who may feel cut off from their friends. Many also face big letdowns as graduations, proms, sports seasons, college visits and other long-planned events are cancelled or postponed.

Here are a few ways you can help you through this difficult time.

1. Stick to a schedule that works with online learning. Set a time to wake up, exercise, shower, get dressed, have breakfast, or whatever they need to start the "school day." If it helps, allow you to sleep in a little later than normal. Like in class, phones should be off while doing schoolwork. Keep the TV off during school hours, too, and limit time watching the news. Plan mini breaks and a 1-hour lunch break.

2. Make dinner a transition time between the "school day" and the evening. Dinner is a great time to gather the whole family together to talk and share a meal. Try fun conversation starters, such as, "My favorite part of today was…" or "Today I am grateful for...". This may be the time your family may choose to observe a quiet moment together. Keep the usual sleep timeroutine.

3. Allow "down time." It's normal for teens to crave more privacy from their family. Space for some quiet time, creative time, music time, or to virtually hang out with friends can help ease any feelings of being isolated from their friends or difficulties with routine-change.

4. Communicate honestly & openly:

   Share information about what is happening in a calm and factual way to help ease their concerns about the virus. Discuss facts about COVID-19 and correct misinformation when you hear it. Reinforce the basics, like the importance of frequent hand washing and avoiding touching their face.

5. Stress that staying home saves lives. Talk about how social distancing is an important way they are helping slow the spread of the virus and protecting those most at risk. Havea strict “no cheating" rule and stress that it is NOT okay to hang out with friends in personor play outdoor sports like basketball and softball.

6. Talk about how they're feeling during the pandemic. Watch for signs of struggling and may need more support, or any signs of increased suicide risk.

7. Stay safely connected

   Reach out virtually. Allow you to stay connected to friends and loved ones during social distancing by phone, text, video chat, or social media. (Remind them to check their privacy settings so they are not posting too much personal information online.) Playing games online with friends can also be relaxing and enjoyable for you. But be sure to agree on screen time during school days.

8. Help others connect. Many teens have expertise in using technology and can teach parents or grandparents how to video chat or use social media.

What about media use?

While limits are still important, it's understandable that under these unusual circumstances, your teen's screen time will likely increase. Work together to come up with a plan that includes both online and offline time. Our Family Media Plan tool can help. Allowing your teen to be a part of making a media plan can help them stick to it.

Support Family & Community

- New responsibilities. Routines have changed, and your family may need extra help in caring for younger children or keeping the house clean.
- Virtual guests. Come up with creative ways to stay connected with family and friends on a regular basis. Try hosting a "virtual dinner" by setting up a laptop or iPad at the table with the invited guests. Or use a video conferencing platform like Zoom to have an online party where everyone can see each other.
- Family projects. Suggest to take the lead in projects that involve the entire family, like organizing family photos or recreating the family's history. This is a great topic for calls to grandparents, who may be able to describe challenging times in the past, and how the family coped with stress.
- Declutter & donate. Clean out your room, the basement, or declutter the garage and prepare items to donate to charity.
- Volunteer within the community. Following social distancing and local regulations, suggest to look online for local opportunities to serve. Show acts of kindness by making someone's day better with a phone call, text, or social media post. Volunteer to help tutor children of neighbors or friends online.

Risky Times for Risky BeHaviors

As schools move to online learning, teens may have more unscheduled time. This can leave them more likely to experiment with risky behaviors. Be sure to talk with your teen about how this is an especially important time to avoid vaping and smoking, for example. Experts warn these habits may harm lung health and immune function, potentially increasing COVID-19 risk.

Mind and body health: find ways to keep their mind and body healthy, such as:

- Go for a walk or a run outside, either by themselves or as a family. Remind them of the social distancing rules and to stay 6 feet away from others.
- Read a book or visit the library online where there are thousands of e-books, audiobooks, and musical recordings. Research new hobbies or skills to learn.
- Do video workouts. Many can be found online, and some park districts are offering access to virtual exercise classes, too.
- Watch movies or TV shows together as a family or virtually with friends.
- Create a video blog of life during the COVID-19 outbreak. Or, start a family journal where each family member can take turns describing the day's happenings.
- Take a virtual tour of a museum, or walk through the Grand Canyon with Google Earth. Challenge your teen to research 10 places they might like to visit someday and show you why.
- Get plenty of sleep!

The COVID-19 pandemic has changed education forever. This is How. (Li & Lalani, 2021)

- The COVID-19 has resulted in schools shut all across the world. Globally, over 1.2 billion children are out of the classroom.
- As a result, education has changed dramatically, with the distinctive rise of e-learning, whereby teaching is undertaken remotely and on digital platforms.

- Research suggests that online learning has been shown to increase retention of information, and take less time, meaning the changes coronavirus have caused might be here to stay.

While countries are at different points in their COVID-19 infection rates, worldwide there are currently more than 1.2 billion children in 186 countries affected by school closures due to the pandemic. In Denmark, children up to the age of 11 are returning to nurseries and schools after initially closing on 12 March, but in South Korea students are responding to roll calls from their teachers online.

With this sudden shift away from the classroom in many parts of the globe, some are wondering whether the adoption of online learning will continue to persist post-pandemic, and how such a shift would impact the worldwide education market.

Even before COVID-19, there was already high growth and adoption in education technology, with global edtech investments reaching US$18.66 billion in 2019 and the overall market for online education projected to reach $350 Billion by 2025. Whether it is language apps, virtual tutoring, video conferencing tools, or online learning software, there has been a significant surge in usage since COVID-19.

In response to significant demand, many online learning platforms are offering free access to their services, including platforms like BYJU'S, a Bangalore-based educational technology and online tutoring firm founded in 2011, which is now the world's most highly valued edtech company. Since announcing free live classes on its Think and Learn app, BYJU's has seen a 200% increase in the number of new students using its product, according to Mrinal Mohit, the company's Chief Operating Officer.

Tencent classroom, meanwhile, has been used extensively since mid-February after the Chinese government instructed a quarter of a billion full-time students to resume their studies through online platforms. This resulted in the largest "online movement" in the history of education with approximately 730,000, or 81% of K-12 students, attending classes via the Tencent K-12 Online School in Wuhan.

Other companies are bolstering capabilities to provide a one-stop shop for teachers and students. For example, Lark, a Singapore-based collaboration suite initially developed by Byte Dance as an internal tool to meet its own exponential growth, began offering teachers and students unlimited video conferencing time, auto-translation capabilities, real-time co-editing of project work, and smart calendar scheduling, amongst other features. To do so quickly and in a time of crisis, Lark ramped up its global server infrastructure and engineering capabilities to ensure reliable connectivity.

Alibaba's distance learning solution, DingTalk, had to prepare for a similar influx: "To support large-scale remote work, the platform tapped Alibaba Cloud to deploy more than 100,000 new cloud servers in just two hours last month – setting a new record for rapid capacity expansion," according to DingTalk CEO, Chen Hang.

Some school districts are forming unique partnerships, like the one between The Los Angeles Unified School District and PBS SoCal/KCET to offer local educational broadcasts, with separate channels focused on different ages, and a range of digital options. Media organizations such as the BBC are also powering virtual learning; Bitesize Daily, launched on 20 April, is offering 14 weeks of curriculum-based learning for kids across the UK with celebrities like Manchester City footballer Sergio Aguero teaching some of the content.

Supplemental Reading:

https://www.deped.gov.ph/wp-content/uploads/2021/02/DO_s2021_001.pdf

https://time.com/5791661/who-coronavirus-pandemic-declaration/

https://www.weforum.org/agenda/2020/06/inventions-coronavirus-covid19-innovation/

https://sciencebusiness.net/covid-19/news/computer-science-versus-covid-19

Lesson 4. The Social Impact of Technology (Immink, 2019)

As human beings, we do not understand exponential or self-amplifying. Particularly the impact on humanity. The dark trends:

1. **Dependency** — leaving our thinking to software and algorithms because it's just so much more convenient and fast.
2. **Confusion** — not knowing if it was the intended human who replied to my emails, or her AI assistant. Or even not knowing if I made my own decision or if I was manipulated by my IDA.
3. **Loss of control** — not having a way of knowing if the AI's anticipation was correct or not, as we could not possibly track the system's logic or even comprehend the workings of a quantum computing-fueled, machine-learning system. In other words, we would need to either trust it completely or not at all, similar to the dilemma that some airplane pilots are already facing with their autopilot systems.
4. **Abdication** — being tempted to leave more tasks to systems that would handle them for us, whether it is coordinating personal schedules, making appointments, or answering simple emails. Then, of course, it would be very likely that we would simply blame the cloud/bot/AI if something went wrong.
5. **Social autism** (we love our screen more than we love people).
6. **Addiction to technology** ("mobile devices are the new cigarettes").
7. **Digital obesity**. Every consumer in developed countries unwittingly ingests an estimated 150 pounds of additives — mostly sugar, yeast, and antioxidants, as well as truly nasty stuff such as MSG. Thus consumers are strung along by cleverly engineering a "need-for-more" so that it becomes very hard to find the exit from that kingdom of endless, happy consumption.

   If this sounds like Facebook or your smartphone, you are getting my drift. The food industry actually calls this capability or crave-ability. In the world of technology, marketers call it magic, stickiness, indispensability, or more benignly, user engagement. Craving and addiction as tech's business model. Think 2020 and imagine billions of hyper connected

consumers becoming digitally obese, hooked on a constant drip of information, media, and data — and their own feedback loops.

8. **Digital feudalism** (winners, i.e. platform winning it all).
9. **Security**, because with virtualization comes decentralizing with many fewer points of physical control.
10. **Software soon eating biology**," and the increasing temptation to virtualize humans via brain-uploading or cyborgism — the dream of many transhumanists.
11. **Forgetting ourselves exponentially** and sleepwalking through digital life, opening the door to a kind of global digital feudalism — where the overlords of technology rule us in ways that are beyond our understanding.
12. **Treating people** in a social security environment just by the numbers, as disembodied data source.
13. **The development of digital egos** as a true copy of ourselves thanks to a combination of fast, cheap, and ultra-powerful tools, including mobile cloud technologies, personalization,voice and image recognition, mood analytics, and sentiment analysis. Eventually, we will be constantly connected to machines, and they are getting better and better at reading our minds. It will lead to the end of free will.

    How our choices will be shaped if what we see and hear about each other is determined purely by algorithms that are designed to make you stay and view ads as long as possible, rather than by people? What if these tools are not publicly controlled, supervised or regulated…? As Taleb said "The difference between technology and slavery is that slaves are fully aware that they are not free"

14. **Software** no longer just "eating the world" but increasingly "cheat the world."
15. Everyone and everything become a data beacon, generating thousands of gigabytes per day, collected, filtered, and analyzed in the cloud by armies of IBM's Watsons and Google's Deep Minds applying their hungry, self-learning global AI brains every second.
16. **Low-cost**, ubiquitous digital technologies have made it possible for us to outsource our thinking, our decisions, and our memories to ever-cheaper mobile devices and the intelligent clouds behind them. These "external brains" are morphing quickly from knowing-me to representing-me to being-me.

17. In its darkest variation, the IoT could be the climax of machine thinking — the most perfect spying operating system (OS) ever devised, the largest real-time surveillance network ever contrived, enforcing total human compliance and killing off all remaining semblance of anonymity.

Sitting ducks

We will become sitting ducks for manipulation and undue influence by anyone who knows how to use the system. As biology gives way to technology, our biological systemswill become increasingly optional, replaceable, and finally even vestigial.

Internalization of Technology

Technology is going internal — separating us from the world, increasingly disconnecting us from human experiences. We are already starting to confuse the magicof the tools with the drug-like effect of constant connectivity, mediation, screenification, simulation, and virtualization. The magic is already becoming manic — addictive, tempting, nudging, demanding — so what will happen when the magic quotient reaches 1,000, when technology becomes infinitely more powerful, cheap, and inseparable from us? At some point in the not-so-distant future we may have to consider the ultimate question: Do we now live inside the machine, or does the machine live inside of us? Datais the new oil: pay or become the content.

The gap

We are facing an enormous gap between what technology can do (the answer seems to be pretty much anything), and what it should do to result in overall human happiness. To safeguard humanity's future, we must invest as much energy in furthering humanity as we do in developing technology. Algorithms can measure or even simulate everything except for what really matters to humans. Misdefining what human flourishing means, will only empower machines.

Humanity is at risk

The risk with that is that we will be losing humanity. Automation is exploding because it's abundantly clear that humans are expensive, slow, and often inefficient, whereas machines are cheap, fast, ultra-efficient, and becoming exponentially more so.

Debug humanity

We are in danger of debugging humanity. Debugging mystery, mistakes, and serendipity, and debugging slow, tedious human behaviors like discussion, pondering, and emotions. Being human is just too cumbersome. Automation is exploding because it's abundantly clear that humans are expensive, slow, and often inefficient, whereas machines are cheap, fast, ultra-efficient, and becoming exponentially more. Will we eventually be a species completely devoid of consciousness, mystery, spirituality, and soul, simply because there's no room for these androrithms in this coming machine age? Zoom forward another ten years, and we may indeed end up 95% automated, hyper connected, virtualized, uber-efficient, and much less human than we could ever imagine today.

Happiness

- Humans seem happiest when they have:
- Pleasure (tasty food, warm baths)
- Engagement (or flow, the absorption within an enjoyed yet challenging activity)
    - Relationships (social ties have turned out to be an extremely reliable indicator of happiness) Meaning (a perceived quest or belonging to something bigger)
- Accomplishments (having realized tangible goals).

In contrast, technology will be of material help in furthering real relationships, or in establishing sense, purpose, or meaning. In fact, quite the opposite may be true, as technology can often be quite corrosive to relationships, as when we obsess with our mobile devices at a family dinner.

Compassion

Compassion — a unique trait connected to happiness. Can you imagine a computer, an app, a robot, or a software product that has compassion?

Hacking happiness?

Happiness cannot be acquired or purchased, and therefore would be impossible to stuff into an app, a bot, or some other machine. Supporting evidence suggests that experiences have a much longer impact on our overall happiness than possessions. Experiences are personal, contextual, timely, and embodied.

However, the key argument of the techno-progressive thinkers is that being happyis just the result of the right kinds of neurons firing at the right time, in the right order. Theyreason that it's all just biology, chemistry, and physics and can thus be understood, learned, and copied completely by computers.

Manipulation

And soon, it will be done via very skillful manipulation of our senses. Computers will try to make us feel happy. They will try to be our friends. And they'll want us to love them.

We will unravel the genetic determinants of key neurotransmitters like serotonin, dopamine, and oxytocin, and be able to manipulate happiness genes — if not serotonin-related 5-HTTLPR then something like it — with precise nanoscale technologies that marry robotics and traditional pharmacology. These "mood bots," once ingested, will travel directly to specific areas of the brain, flip on genes, and manually turn up or down our happiness set point, coloring the way we experience circumstances around us.

Mantal arms race

In a review of "The science of selling" predicting an arms race between mindfulness and (sales) manipulation. Compassion and happiness, like consciousness, simply do not exist in mere biological or chemical terms but in the holistic interplay of everything that is human. Machines or software are unlikely to ever attain these states, even if they quickly become better at simulating

them to some extent.

Attempts at first defining and then programming a human characteristic such as compassion, or something as mysterious as consciousness, seem like a far-fetched and unrealizable concept in the foreseeable future. But then again, is the real danger that a great simulation (which will include AR and VR) will quite possibly be "good enough" for most of us?

Can your program happiness?

Happiness cannot be programmed into machines, automated, or sold. It cannot be copied, codified, or deep-learned. It needs to emanate from and grow within us, and in between us, and technology is here to help us — as a tool. We are a species that uses technology, not a species that is destined to be (come) technology.

Human rights

- He suggests five new human rights for the Digital Age
- The right to remain natural, i.e. biological — we must have the choice to exist in an augmented state.
- The right to be inefficient if and where it defines our basic humanness
- We must have the choice to be slower than technology.
- The right to disconnect — we must retain the right to switch off connectivity, to "go dark" on the network, and to pause communications, tracking, and monitoring.
- The right to be anonymous — In this coming hyper connected world, we should still have the option of not being identified and tracked,

And some rules:

- ✓ We shall not empower machines to empower themselves, and thereby circumvent human control.
- ✓ We shall not seek to minimize human flaws just to make a better fit with technology.
- ✓ We shall not attempt to abolish mistakes, mystery, accidents, and chance by using technology to predict or prevent them, and we shall not strive to make everything explicit just because technology may make it feasible to do so.
- ✓ We shall not create, engineer, or distribute any technology with the primary goal of generating addiction to it.
- ✓ We shall not confuse a clean algorithm for an accurate picture of human reality
- ✓ We must not pursue efficiency over humanity

Could or should?

What can we do about technology taking over where it should not? How can we protect ourselves from merely becoming the objects of bot-fueled hyper-efficiency, feedinga giant AI that in turn dictates our lives and tells us what we can no longer do? In the very near future, it will no longer be about whether technology can do something (the answer will almost always be yes) but whether it should do. He is concerned that we may soon become completely useless without technology — slow, incomplete, dumb, deskilled, lazy,and obese.

Erosion of humanity

Imagine what would happen if we continued to chip away at and ultimately erode quintessential human qualities such as privacy, mystery, anonymity, emotions, spontaneity, surprise, intuition, imagination, and spirituality — just so that we can keep upwith the machines.

Stewardship

We must become much better stewards of humanity. Every single business leader, technology pioneer, and public official needs to accept and act upon his/her responsibility for shaping the future of humanity. Technology has no ethics, yet a society without ethicsis doomed. Exponential technologies often morph rapidly from magic to manic to toxic — achieving a balance

is essential.

We need to teach both STEM and CORE (compassion, originality, reciprocity, and empathy) skills. Technology and humanity must both be on the curriculum; indeed scienceand philosophy belong in the same classroom.

We need to retain a clear distinction between what is real and what is a copy or a simulation. Total connectivity, thinking machines, the smart cloud, and cognitive computing are our inevitable future, yet we should not abandon the distinction between simulation and being, between computation and sentience, between machine-kind and mankind.

We need to start asking why and who, not just if or how.

We should not let Silicon Valley, technologists, the military, or investors become mission control for humanity — no matter what country they are in. Those who fund, create, and sell exponential technologies are unlikely to be the ones who will want to curbtheir power or scale of potential applications.

Seven questions

- When we evaluate exponential technologies, we should ask seven essential questions:
- Will this technology inadvertently or by design diminish humanity?
- Will this technology further true human happiness?
- Does this technology have any unintended and potentially disastrous side effects?
- Will this technology give too much authority to itself or to other algorithms, bots, and machines?
- Will this technology enable us to transcend it, i.e. go beyond itself, or will it make us dependent on it?
- Will humans need to be materially changed or augmented to actually use this technology?
- Will this technology be openly available, or will it be proprietary?

Supplemental Readings:

https://www.weforum.org/agenda/2020/06/inventions-coronavirus-covid19-innovation/

https://www.weforum.org/agenda/2020/04/10-technology-trends-coronavirus-covid19-pandemic-robotics-telehealth/

Assessment Task 7-1

Application of Computers in Science and Research during the COVID-19 Pandemic

Objective:

Evaluate students' understanding of how computers have been utilized in scientific research, particularly during the COVID-19 pandemic, and their ability to analyze the impact of technology on advancing scientific knowledge.

Instructions:

1. **Background Research (10 points):**
   - Provide a brief overview of the role of computers in scientific research.
   - Explore key concepts such as computational modeling, data analysis, and simulation.
2. **Application of Computers in COVID-19 Research (20 points):**
   - Investigate specific instances where computers have played a crucial role in COVID-19 research.
   - Discuss how computational methods contributed to vaccine development, epidemiological modeling, and understanding the virus.
3. **Impact Assessment (15 points):**
   - Evaluate the impact of computer applications on the speed and efficiency of COVID-19 research.
   - Discuss how technology facilitated collaboration among researchers and accelerated the dissemination of findings.
4. **Challenges and Solutions (15 points):**
   - Identify challenges faced in utilizing computers for COVID-19 research.
   - Propose potential solutions or improvements to overcome these challenges.
5. **Case Study Analysis (20 points):**
   - Choose a specific case study where computers played a pivotal role in COVID-19 research (e.g., drug discovery, genomic analysis).
   - Analyze the methodology, tools, and outcomes of the study.
6. **Innovative Applications (10 points)**:
   - Suggest innovative ways computers could be further utilized in scientific research beyond the pandemic.
   - Discuss potential advancements in technology that may revolutionize scientific inquiry.

This assessment task aims to assess research skills, critical analysis, and the ability to articulate the significance of computer applications in scientific research during the COVID-19 pandemic.

Assessment Task 7-2

Complete the table: (15 pts)

Country	Research/Invention/Solutions or Software	Functions, Purpose. Impact	Reference/s
Philippines			
United States			
European Country			
Other Asian Country			
Other			

*Or your professor can create his/her own set of questions for this AT.

Summary

The integration of digital technology into pandemic policy and response could be one of several characteristic features of countries that have flattened their COVID-19 incidence curves and maintained low mortality rates. In the race to contain the spread of a highly transmissible virus, countries that have quickly deployed digital technologies to facilitate planning, surveillance, testing, contact tracing, quarantine, and clinical management have remained front-runners in managing disease burden. The comprehensive responses of countries that have been successful at containment and mitigation can provide insight to other countries that are still facing a surge of cases.

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MODULE 8
Genes

Introduction

According to Bartee (2019), “genetics” is the study of how traits are inherited. A trait is defined as a variation in the physical appearance of a heritable characteristic. It seeks to understand how traits are passed from generation to generation.

Genes are specific sequences of nucleotides that code for particular proteins. Through the processes of meiosis and sexual reproduction, genes are transmitted from one generation to the next. Augustinian monk Gregor Mendel developed the science of genetics. Mendel performed his experiments in the 1860s and 1870s, but the scientific community did not accept his work until early in the twentieth century. Because the principles established by Mendel form the basis for genetics, the science is often referred to as Mendelian genetics. Mendel believed that factors pass from parents to their offspring, but he did not know of the existence of DNA. Modern scientists accept that genes are composed of segments of DNA molecules that control discrete hereditary characteristics (Cliffnotes.com, 2015).

Learning Outcomes

At the end of this module, students should be able to:

1. Discuss the meaning of genes
2. Differentiate gene therapy for stem cell therapy
3. Explain the importance of gene therapy in the future
4. Demonstrate the process of gene therapy
5. Assess the issue’s potential benefits and detriments to global health.

Lesson 1. Definition of Gene

Charles Darwin used the term Gemmule to describe a microscopic unit of inheritance. This came to be known as Chromosomes. Chromosomes, however, were first observed during cell division by Wilhelm Hofmeister as early as 1848 (Mandal, 2010).

Wilhelm Roux in 1883 speculated that chromosomes are the carriers of inheritance. Danish botanist Wilhelm Johannsen coined the word "gene" ("gen" in Danish and German) in 1909 to describe these fundamental physical and functional units of heredity. William Bateson in 1905 coined the term genetics from the word gene (Mandal, 2010).

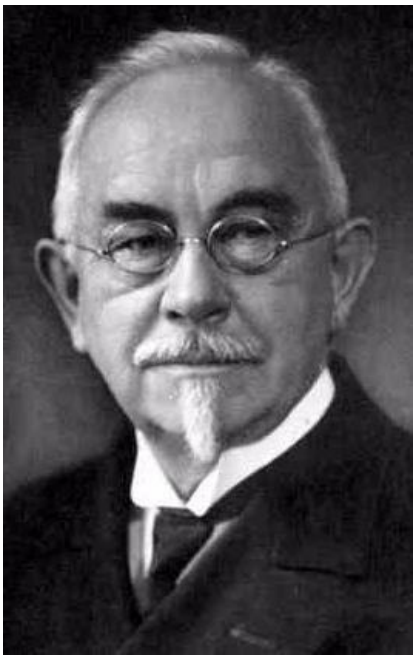

Figure 8.1 Wilhelm Johannsen
(Sutori.com, 2016)

According to Medicalnews.com (2020), a gene is a basic unit of heredity in a living organism. Genes come from our parents. We may inherit our physical traits and the likelihood of getting certain diseases and conditions from a parent. Genes contain the data needed to build and maintain cells and pass genetic information to offspring. Each cell contains two sets of chromosomes: One set comes from the mother and the other comes from the father. The male sperm and the female egg carries a single set of 23 chromosomes each, including 22 autosomes plus an X or Y sex chromosome. A female inherits an X chromosome from each parent, but a male inherits an X chromosome from their mother and a Y chromosome from their father.

An international research effort called the Human Genome Project, which worked to determine the sequence of the human genome and identify the genes that it contains, estimated that humans have between 20,000 and 25,000 genes (Medlineplus.gov, 2020).

Every person has two copies of each gene, one inherited from each parent. Most genes are the same in all people, but a small number of genes (less than 1 percent of the total) are slightly different between people. Alleles are forms of the same gene with small differences in

their sequence of DNA bases. These small differences contribute to each person's unique physical features (Medlineplus.gov, 2020).

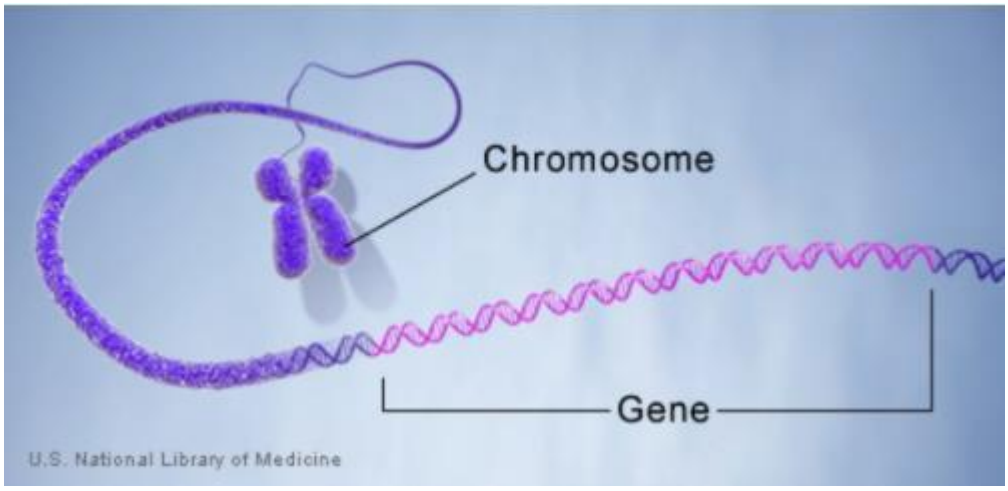

Figure 8.2. Gene

Source:(Medlineplus.gov,2020)

Lesson 2. Difference between Gene Therapy and Stem CellTherapy

A. The Aspect of GeneTherapy

According to Serafica, Pawilen, Casilib & Alata (2018), medical science has detected many human diseases related to defective genes. These types of diseases are not curable by traditional methods like taking readily available medicines. Gene therapy is a potential method to either treat or cure genetic-related human illnesses.

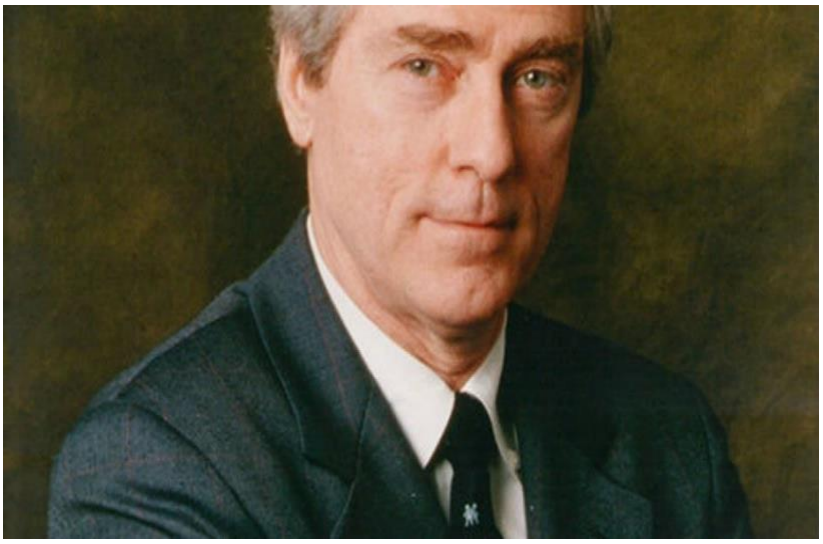

Figure 8.3. William French Anderson "The Father of Gene Therapy" (Slideshare.net, 2011)

In 2015, a team of researchers at the Harvard Medical School and the Boston Children's Hospital stated that they were able to restore basic hearing in genetically deaf mice using gene therapy. The Boston Children's Hospital research team also reported that they have restored a higher level of hearing—down to 25 decibels which is actually equivalent to a whisper. They used an improved gene therapy vector developed at the Massachusetts Eye and Ear that was identified as "Anc80" which enables the transfer of genes to the inaccessible outer hair cells when introduced into the cochlea (Fliesler, 2017 as cited in Serafica et al., 2018).

Human gene therapy was actually first realized in 1971 when the first recombinant DNA experiments were planned. It can be simple viewed as insertion foreign DNA into a patient's tissue that hope to successfully eradicate the targeted disease. It was actually inspired by the success of recombinant DNA technology which occurred over the last 20 years. Without a doubt, gene therapy is the most promising yet possibly unfavorable medical field being studied.

The Basic Process

There are several approaches to gene therapy. These are the following (Fliesler, 2017, as cited in Serafica et al., 2018):

- Replacement of mutated gene that causes disease with a healthy copy of the gene
- Inactivation of a mutated gene that is functioning improperly
- Introducing a new gene into the body to help fight disease

In general, a gene cannot be directly inserted into a human gene or cell. A gene is inserted into another gene using a carrier or vector. At present, the most common type of vectors are viruses that have been genetically changed to carry normal human DNA. Viruses have evolved a way of encapsulating and transporting their genes to human cells in a pathogenic manner (Science Daily, 2017 as cited in Serafica et al., 2018).

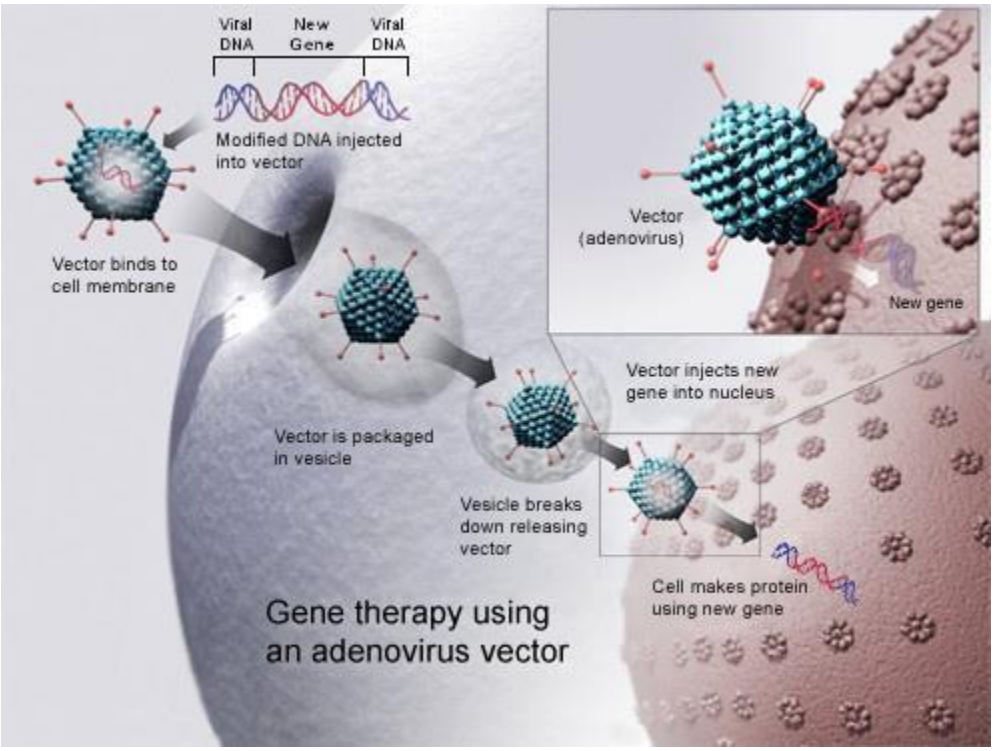

Figure 8.4. Gene Therapy

Source: (Slideshare.net, 2011)

Two Types of Gene Therapy

The idea of gene therapy is based on correcting a disease at its root fixing the abnormal genes that appear to lead to certain diseases.

There are essentially two forms of gene therapy. One of which is called somatic gene therapy.

Somatic gene therapy involves the manipulation of genes in cells that will be helpful to the patient but not inherited to the next generation (Nimsergern, 1988 as cited in Serafica et al., 2018).

The other form of gene therapy is called **Germ-line gene therapy** which involves the genetic modification of germ cells or the origin cells that will pass the change on to the next generation (Your Genome, 2017 as cited in Serafica et al., 2018).

B. Stem Cell Therapy

Therapy Stem cells are mother cells that have the potential to become any type of cell in the body. One of the main characteristics of stem cells is their ability to self-renew or multiply while maintaining the potential to develop into other types of cells. Stem cells can become cells of the blood, heart, bones, skin, muscles, and brain, among others. There are different sources of stem cells but all types of stem cells have the same capacity to develop into multiple types of cells (Serafica et al., 2018).

Stem cells are derived from different sources. Two of which are embryonic and somatic stem cells.

The **embryonic stem cells** are derived from a four- or five-day-old human embryo that is in the blastocyst phase of development. The embryos are usually extras that have been created in IVF (in vitro fertilization) clinics where several eggs are fertilized in a test tube then implanted into a woman (Crosta, 2013 as cited in Serafica et al., 2018).

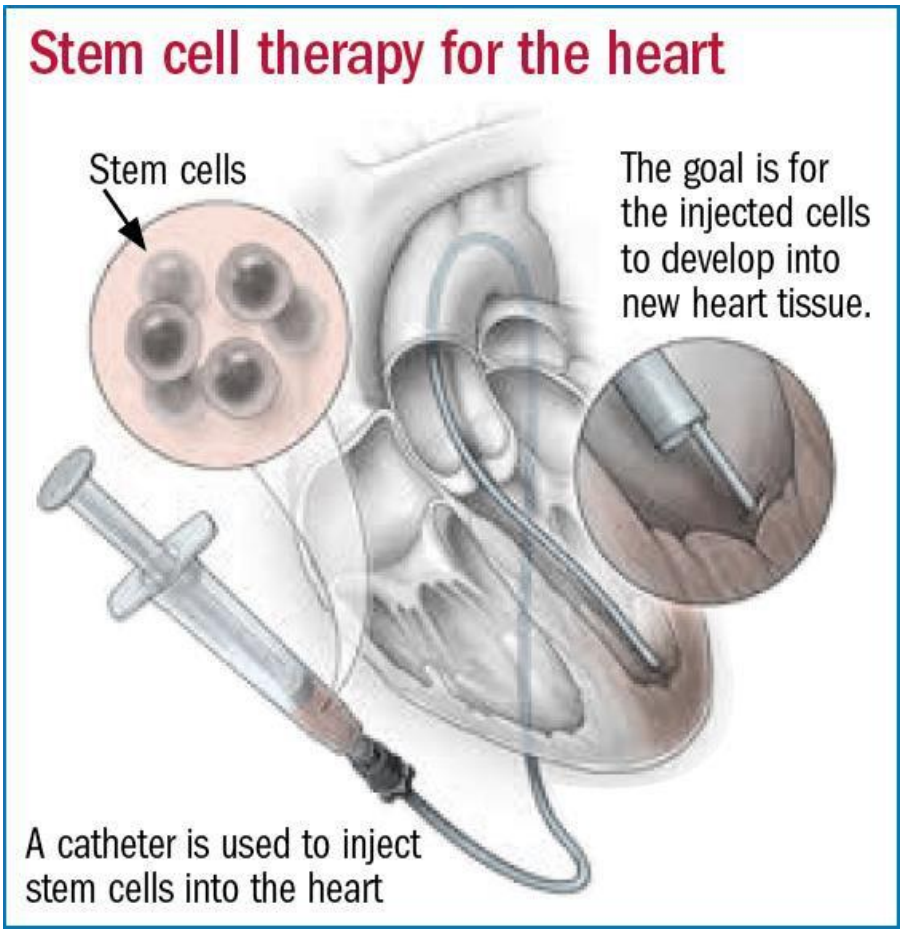

Figure 8.5 Stem cell to repair heart damage

Source: (health.harvard.edu, 2020)

The **somatic stem cells** are cells that exist throughout the body after embryonic development and are found inside different types of tissue. These stem cells have been found in tissues such as the brain, bone marrow, blood, blood vessels, skeletal muscles, skin, and the liver. They remain in a non-dividing state for years until activated by disease or tissue injury. These stem cells can divide or self-renew indefinitely, enabling them to generate a range of cell types from the originating organ or even regenerate the entire original organ. It is generally thought that adult or somatic stem cells are limited in their ability to differentiate based on their tissue of origin, but there is some evidence to suggest that they can differentiate to become other cell types (Crosta, 2013 as cited in Serafica et al., 2018).

Cell therapy is the transfer of cells into a patient in order to improve or even cure a disease. The origin of the cells depends on the treatment. The transplanted cells are often a type of adult

or fetal stem cell which have the ability to divide and self-renew as well as provide cells that mature into the relevant specialized cells of the tissue. A common form of cell therapy is bone marrow transplantation, which has been performed for over 40 years (Futurelearn.com, n.d.).

Several investigative protocols of cell therapy involve the transfer of adult T lymphocytes that are genetically modified to increase their immune potency and can self-renew and kill the disease-causing cells (Futurelearn.com, n.d.).

Stem cells from umbilical cord blood and other tissues are being developed to treat many genetic diseases and some acquired diseases. Stem cells from the fetus may be better than adult stem cells; they divide more and differentiate better into the definitive cell type, for example fetal liver stem cells produce much more bone than adult liver stem cells (Futurelearn.com, n.d.).

C. How are gene therapy and cell therapy related?

Both approaches have the potential to treat the underlying cause of both genetic and acquired diseases by replacing the missing proteins or cells, and they may suppress expression of proteins that are toxic to cells (Futurelearn.com, n.d.).

Whereas gene therapy involves the transfer of genetic material into the appropriate cells, cell therapy is the transfer of cells to a patient. Gene therapy involves the transfer of genetic material usually in a carrier or vector, and the uptake of the gene into the appropriate cells of the body. Cell therapy involves the transfer of cells with the relevant function into the patient (Futurelearn.com, n.d.).

Risks of cell therapy also include the loss of tight control over cell division in the stem cells. Theoretically, the transplanted stem cells may gain a growth advantage and progress to a type of cancer or teratomas, but this is only rarely seen in stem cell transplantation, and has not yet been seen in fetal stem cell transplantation (Futurelearn.com, n.d.).

Since each therapy has its potential risks, patients are strongly encouraged to ask questions of their investigators and clinicians until they fully understand the risks. Stem cells can theoretically improve the disease condition for as long as the modified stem cells remain live, potentially the

entire lifetime of the patient (Futurelearn.com, n.d.).

Lesson 3. Importance of Gene Therapy in the Future

Eventually, gene therapy will become a staple of 21st century medicine. But some experts say society will be better served if medical researchers proceed more slowly and prudently (McCain, 2005).

In its current manifestation, gene therapy is an elegant concept crudely executed. That's not an indictment — that's just the way it is for an extraordinarily complicated technology still in its infancy. After all, it has been only 5 years since the concept of gene therapy was convincingly demonstrated to provide, if not a cure, then at least a long-term therapeutic effect for X-linked severe combined immunodeficiency (X-SCID) disease (McCain, 2005).

A "bionic chip"

As previously discussed, genes can be delivered into cells in different ways. The selection of a gene delivery system depends on the target cell, the duration of gene expression required for therapeutic effect, and the size of the piece of DNA to be used in the gene therapy (Medicinet.com, 2020).

Genes can be carried into cells by **viruses**. Viral vectors or carriers take advantage of the natural ability of a virus to enter a cell and deliver genetic material to the nucleus of the cell that contains its DNA. In developing virus carriers, the DNA coding for some or all of the normal genes of the virus to be used as a carrier are removed and replaced with a treatment gene. Most of these virus carriers are engineered so that they are able to enter cells, but they cannot reproduce themselves and so are innocuous (Medicinet.com, 2020).

Genes can also be delivered within tiny synthetic "envelopes" of fat molecules. Cell membranes contain a very high concentration of fat molecules. The fat molecule "envelope" can carry the therapeutic gene into the cell by being admitted through the cell membrane as if it were one of its own molecules (Medicinet.com, 2020).

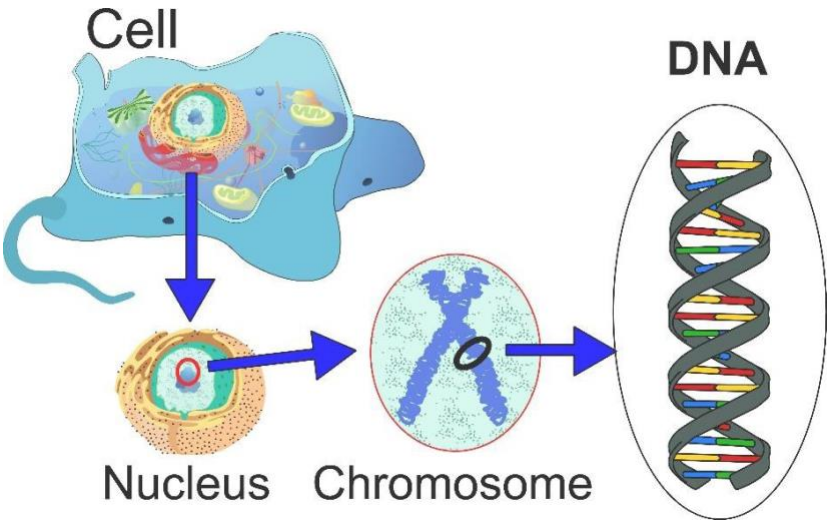

Figure 8.6. DNA Basis

Source: (genetics.thetech.org,2018)

Genes can also gain entrance into cells when an electrical charge is applied to the cell to create tiny openings in the membrane that surrounds a cell. This technique is called electroporation (Medicinet.com, 2020).

Now, the is a new way on how to deliver the genes into the cells. A **"bionic chip"** has been developed to help gene therapists using electroporation to slip fragments of DNA into cells. Electroporation was originally a hit-or-miss technique because there was no way to determine how much of an electrical jolt it took to open the cell membrane (Medicinet.com, 2020).

The "bionic chip" solves this problem. It contains a single living cell embedded in a tiny silicon circuit. The cell acts as a diode, or electrical gate. When it is hit with just the right charge, the cell membrane opens, allowing the electricity to pass from the top to the bottom of the bionic chip. By recording what voltage caused this phenomenon to occur, it is now possible to determine precisely how much electricity it takes to pry open different types of cells (Medicinet.com, 2020).

Route of administration of gene therapy

The choice of route for gene therapy depends on the tissue to be treated and the mechanism by which the therapeutic gene exerts its effect. Gene therapy for cystic fibrosis, a disease which effects cells within the lung and airway, may be inhaled. Most genes designed to treat cancer are injected directly into the tumor. Proteins such as factor VIII or IX for hemophilia are also being introduced directly into target tissue (the liver) (Medicinet.com, 2020).

The Potential Power of Gene Therapy

Most gene therapy for diseases such as cystic fibrosis and hemophilia has been designed only to ease, not to cure, the disease. However, the delivery of functional copies of genes provides a potential method to correct a disease at its most basic level (Medicinet.com, 2020). Gene therapies also holds the potential to provide "patient- friendly" treatment regimens for a variety of diseases. Today, many patients with hemophilia and diabetes must have repeated injections in order to manage their disease because proteins exist in the blood stream for a limited period of time before they are degraded or eliminated. Since DNA is more stable and functions inside the cell, the delivery of genes may result in longer-term expression of the necessary proteins (Medicinet.com, 2020). Because of its accuracy, gene therapy has the potential to eliminatecancer cells without damaging normal, healthy tissue. Furthermore, cancer gene therapies may provide alternatives when a disease does not respond to other older treatments (Medicinet.com, 2020).

The potential of gene therapy is great but, compared to its promise, the results to date are still quite limited. However, the benefits of gene therapy are believed to be on the near horizon. Gene therapy is one of the hottest areas of medical research today. (And gene therapy companies have been among the hottest in the stock market.) (Medicinet.com, 2020). The remarkable advances in genetics, including the human genome project, have opened new doors for the exploration of gene therapy. New technologies are needed to speed the progress of gene therapy. As these new technologies such as the "bionic chip" arrive, we believe that, without a doubt, gene

therapy will play an increasingly important and prominent part in medicine in the decades to come (Medicinet.com, 2020).

Lesson 4. Role of Genes in Heredity

According to Prieto, Vega, Felipe & Meneses (2019), heredity is the passing on of traits from parents to their offspring, either through asexual reproduction or sexual reproduction, the offspring cells or organisms acquire the genetic information of their parents, through heredity, variations between individuals can accumulate and cause species to evoke by natural selection. The study of heredity is the biology of genetics.

Prieto (2019) added that the heredity information is contained or within the genes, located in the chromosomes of each cell. An inherited trait can be determined by one or by many genes and a single gene can influence more than one trait. A human cell contains many thousands of different genes in the nucleus. Heredity factors known as genes are thought to exist as discrete portions (known as loci) of chromosomes. The term "discrete" refers to the concept that genes are always located at the same point or (locus) on a chromosome.

It is believed that pairs of homologous chromosomes contain linear, matching arrangements of genes exerting parallel control over the same traits. Pairs of genes that exercise such parallel control over the same traits are known as alleles, which are of two kinds: dominant and recessive. Genetic characteristics can be extremely complex and may require the actions of several separate genes to be expressed in the adult organism. An example of traits that require more than a single pair of genes to be expressed are human height and human color. Other traits may be controlled by a single pair of genes. Some of these traits may have extreme phenotypes, such as albinism (lack of skin pigment) in humans (Prieto, 2019).

Common genetic disorder

- down syndrome
- sickle cell disease
- phenylketonuria (PKU)
- hemophilia

Each of our gene codes contains a unique protein and each protein has a separate and distinct function to perform in the cell (Prieto, 2019).

Lesson 5. Effect of Genes in Our Environment

According to Prieto (2019), genes are inherited, but their expression can be modified by interactions with the environment. Genetic traits are determined largely through the precise information found in the cell's gene structure. A variety of factors in the environment can actually alter the effects of a particular gene.

Example:

- Effect of light or chlorophyll production — although most plants have the genetic ability to produce chlorophyll, they will do this only in the presence of light. Without light. These plants produce only a light-yellow pigment and therefore appear pale and sickly until they are exposed to sunlight. After a few days of exposure to sunlight, the chlorophyll production mechanism is enabled and green color returns.

Cells store and use coded information. The genetic information stored in DNA is used to direct the synthesis of the thousands of proteins that each cell requires. The cell contains many thousands of such codes in its chromosomes. Each strand of DNA in the chromosome has the potential to provide the complete chemical code for the manufacture of at least one complete protein. These proteins are highly specific and they result in the expression of some specific trait or portion of a trait in the living cell and, consequently, in the organism of which they are part (Prieto, 2019).

Lesson 6. Genetically Modified Organisms (GMO)

A. Definition of GMO

Genetically modified organisms (GMOs) are living organisms whose genetic material has been artificially manipulated in a laboratory through genetic engineering. This creates combinations of plant, animal, bacteria, and virus genes that do not occur in nature or through traditional crossbreeding methods (Nongmoproject.org 2016).

Most GMOs have been engineered to withstand the direct application of herbicide and/or to produce an insecticide. However, new technologies are now being used to artificially develop other traits in plants, such as a resistance to browning in apples, and to create new organisms using synthetic biology. Despite biotech industry promises, thereis no evidence that any of

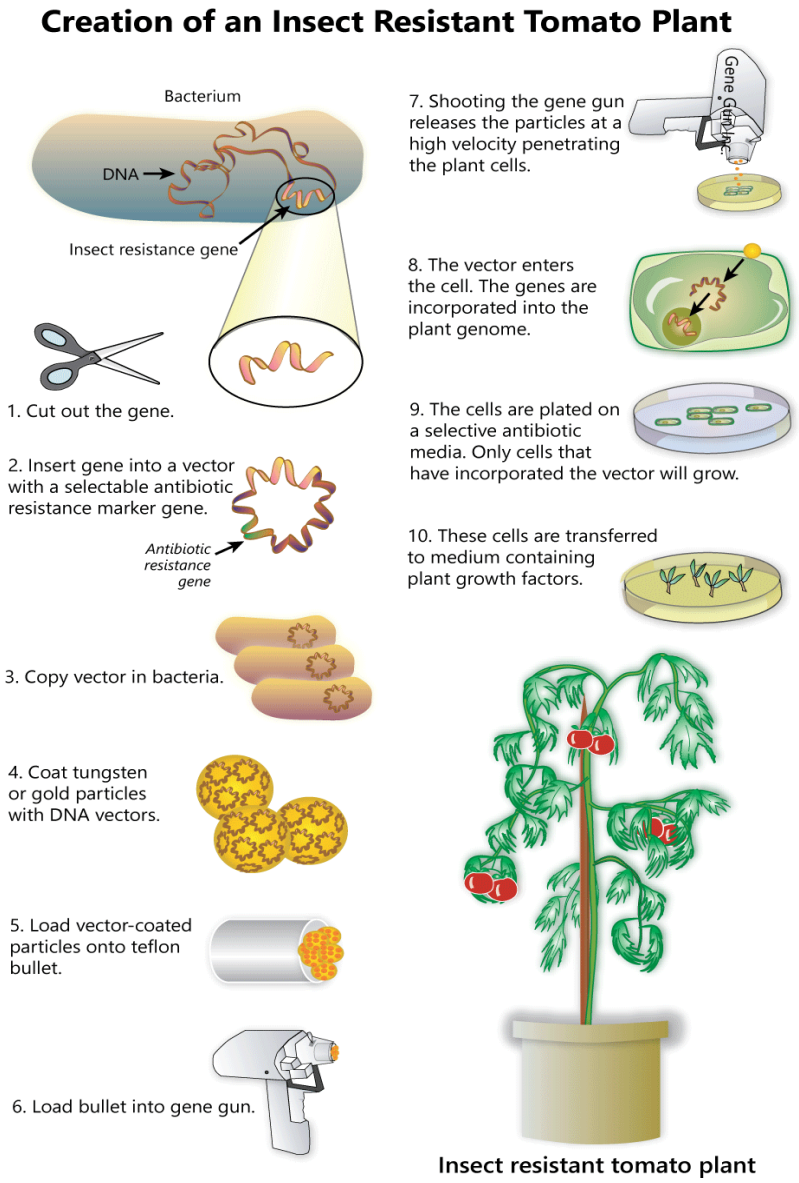

Figure 8.7. Creation of an insect resistant tomato plant

Source: (sbi3ubiotechnology.weebly.com, n.d.)

the GMOs currently on the market offer increased yield, drought tolerance, enhanced nutrition, or any other consumer benefit (Nongmoproject.org 2016).

B. Potential Environmental and Human health risks caused by GMOs

1. Human Health

As the prevalence of genetically modified organisms (GMOs) continues to rise, there has been an increasing public interest for information concerning the safety of these products. Concerns generally focus on how the GMO may affect the environment or how it may affect the consumer. One specific concern is the possibility for GMOs to negatively affect human health. This could result from differences in nutritional content, allergic response, or undesired side effects such as toxicity, organ damage, or gene transfer. To address these concerns, there havebeen over 100 research studies comparing the effects of traditional food to genetically modified food, the results of which have been reviewed in various journals. How these results affect regulation can be found through The Center for Environmental Risk Assessment, which hosts a GM Crop Database that can be searched by the public to find GMO crop history, style of modification, and regulation across the world. Though knowing who to trust and what to believe regarding this topic is an ongoing battle, major health groups, including the American Medical Association and World Health Organization, have concluded from the research of independent groups worldwide that genetically modified foods are safe for consumers. Regarding toxicity, thisincludes any dangers related to organ health, mutations, pregnancy and offspring, and potential for transfer of genes to the consumer (Medicalnewstoday.com, 2020).

GMO toxicity: fears and scientific analysis

After genetically modified foods were introduced in the United States a few decades ago, people independently reported toxic effects caused by GMOs. One example is an anti-GMO advocacy group called the Institute for Responsible Technology (IRT), which reported that rats fed a diet containing a GMO potato had virtually every organ system adversely affected after just ten days of feeding. The IRT stated that the toxicity was the result of genetic modification

techniques and not a specific case for that particular potato. They claimed the process of making

the GMO caused it to be toxic and thus all GMOs were high risk for toxicity (Medicalnewstoday.com, 2020).

Scientists across the U.S. and the rest of the world have sought to rigorously test the assertions of the IRT and others to uncover any possible toxicity caused by GMOs. To this end, many different types of modifications in various crops have been tested, and the studies have found no evidence that GMOs cause organ toxicity or other adverse health effects. An example of this research is a study carried out on a type of GMO potato that was genetically modified to contain the bar gene. The product of the bar gene is an enzyme that can detoxify herbicides and thus protects the potato from herbicidal treatment (Medicalnewstoday.com, 2020).

In order to see if this GMO potato would have adverse effects on consumer health like those claimed by the IRT, a group of scientists at the National Institute of Toxicological Research in Seoul, Korea fed rats diets containing either GMO potato or non-GMO potato. For each diet, they tracked male and female rats. To carefully analyze the rats' health, a histopathological examination of tissues and organs was conducted after the rats died. Histopathology is the examination of organs for disease at the microscopic level (think pathologist doing a biopsy). Histopathological examinations of the reproductive organs, liver, kidneys, and spleen showed no differences between GMO-eating and non-GMO-eating animals (Medicalnewstoday.com, 2020).

Three years earlier, a separate group had found the same results for a GMO tomato and a GMO sweet pepper. These researchers had split rats into four diet groups: non-GMO tomato, GMO tomato, non-GMO sweet pepper, and GMO sweet pepper. They fed the rats over 7,000 times the average human daily consumption of either GMO or non-GMO tomato or sweet pepper for 30 days and monitored their overall health. Finally, they carried out histopathology and again found no differences in the stomach, liver, heart, kidney, spleen, or reproductive organs of GMO versus non-GMO fed rats. Despite massive ingestion of GMO potato, tomato, or sweet pepper, these studies demonstrated no differences in the vitality or health of the animals, even at the microscopic level (Medicalnewstoday.com, 2020).

Experiments like these on humans would be completely unethical. Fortunately, prior to these studies years of work have demonstrated that rodents, like mice and rats, are acceptable models for humans, meaning rodent responses to drugs, chemicals, and foods can predict human

response. Rat feeding studies like these, in which rats are fed a potential toxic item and monitored for adverse effects, are considered both specific and sensitive for monitoring toxicity of foods and widely used in the food regulation industry (Medicalnewstoday.com, 2020).

The test of time: GMOs and their effect on our offspring

To discern whether GMO crops affect fertility or embryos during gestation, a group from South Dakota State University again turned to studies on rats. In this case, the rats were eating a type of GMO corn, more commonly known as Bt corn. Bt stands for Bacillus thuringiensis, a microbe that produces insecticidal endotoxin and has been used as a topical pesticide against insects since 1961. To allow corn to directly generate this endotoxin, scientists introduced a gene from Bt into the genetic material (DNA) of corn (Medicalnewstoday.com, 2020).

To address buildup of toxicity over time, this group monitored the GMO-eatingrats not only for the lifetime of one generation, but also three additional generations. For each generation, they tracked the fertility of parents and compared the health of the embryos from parents that ate Bt corn to those with parents that did not. Toxic effects can arise in many places and in many ways, but some organs are more susceptible to damage than others, and monitoring them is a good readout for other difficult-to-see effects. Testes are considered a particularly sensitive organ for toxicity tests because of the high degree of cell divisions and thus high susceptibility to cellular or molecular toxins. To examine the effect of Bt corn on testicular health, the researchers tracked testicular development in fetal, postnatal, pubertal, and adult rats for all four generations. The group found no change in testicular health or litter sizes in any generation. Likewise, ingestion by pregnant mothers had no effect on fetal, postnatal, pubertal, or adult testicular development of her offspring (Medicalnewstoday.com, 2020).

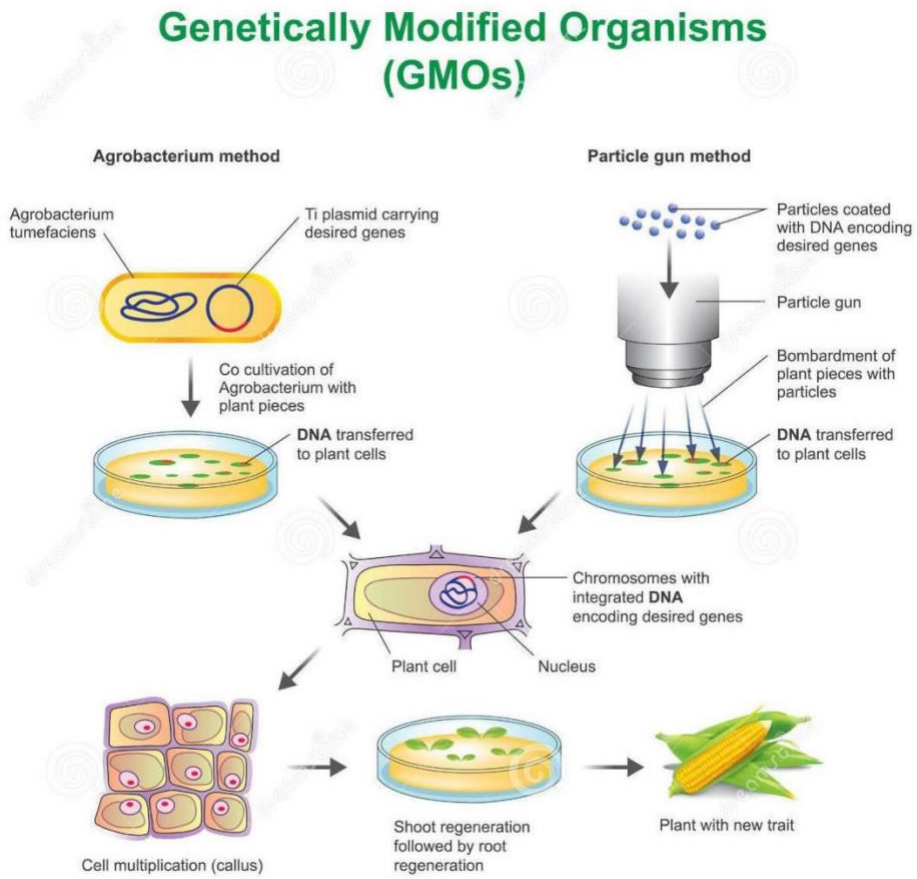

Figure 8.8 Genetically Modified Organisms (GMOs)

Source: (dreamstime.com, n.d.)

Other groups have monitored toxicity over time as well. For example, the group studying the bar GMO potato also wanted to see if organs and reproductive health were sensitive to GMOs

over long exposure times. To do this, they examined the fertility and gestation periods of GMO-eating mothers compared to non-GMO-eating mothers for five generations. They tracked animal body weight, bone, eye, and thymus development, and general retardation. Like the studies on Bt corn, in all cases, they found no significant differences between the GMO potato and non-GMO potato diets, suggesting that there is no buildup or inheritance of toxicity, even over multiple generations (Medicalnewstoday.com, 2020).

A strong argument for GMO health safety

After more than 20 years of monitoring by countries and researchers around the world, many of the suspicions surrounding the effects of GMOs on organ health, our offspring, and our DNA have been addressed and tested. In the data discussed above, alongside many more studies not mentioned here, GMOs have been found to exhibit no toxicity, in one generation or across many. Though each new product will require careful analysis and assessment of safety, itappears that GMOs as a class are no more likely to be harmful than traditionally bred and grownfood sources (Medicalnewstoday.com, 2020).

2. The Environmental Impact of GMOs

According to Royalsociety.org (2016), crops do not damage the environment simply because they are GM. Some farming practices, such as the overuse of herbicides resulting in the excessive eradication of wild plants from farmland have been shown to harm the environment. These problems are similar for non-GM and GM crops.

In a large farm scale evaluation of herbicide tolerant GM crops conducted in the UK between 1999 and 2006 it was shown that when weed control is particularly effective insect biodiversity is reduced. It did not matter whether or not the crop was GM- the important factor was how many weeds remained in the crop. Damage to wildlife can be reduced if a small amount of agricultural land is set aside for biodiversity (Royalsociety,org, 2016).

A related issue is the growing problem of weeds becoming resistant to herbicides, due to

the overuse of those herbicides. Herbicide tolerant crops, whether GM or non-GM, can cause this

problem because repeated growth of the same herbicide tolerant crop involves repeated use of the same herbicide. One solution is the rotation of crops resistant to different herbicides, or rotation of herbicide use with use of other weed control strategies (Royalsociety,org, 2016).

The use of GM crops resistant to insects through introduction of the gene for Bt toxin has environmental benefits. For example, GM insect resistant cotton has substantially reduced the application of more environmentally damaging insecticides, with consequent environmental benefits and health benefits for cotton farmers (Royalsociety,org, 2016).

However, just like herbicide resistant weeds, insect pests can develop resistance to insecticides whether they are produced in the crop itself by GM, or sprayed onto the crop. This problem is less frequent if a rotation of different insect control procedures is used (Royal society, org, 2016).

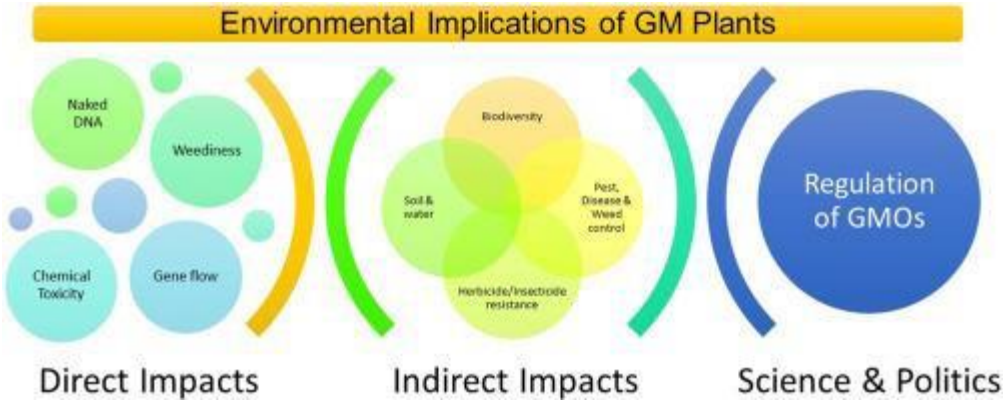

Figure 8.9 .Environmental implications of GM plants

Source: (Tsatsakis, Nawaz, Kouretas, Balias, Savolainen, Tutelyan, 2017).

Assessment Task 8

1. What are the advantages and disadvantages of GMO to human life?
2. What are the advantages and disadvantages of GMO to the environment?
3. Explain the two types of gene therapy
4. How gene therapy differs from stem cell therapy?
5. Should gene therapy be limited to medical concerns only or could it be used for aesthetic purposes? Explain your answer.

Summary

According to Yourgenome.org. (2015), genes are small sections of DNA within the genome that code for proteins. They contain the instructions for our individual characteristics – like eye and hair color. A gene is a small section of DNA that contains the instructions for a specific molecule, usually a protein. The purpose of genes is to store information. Each gene contains the information required to build specific proteins needed in an organism.

Gene therapy is a method that may treat or cure genetic-related human illnesses. There are two forms of gene therapy. One is somatic gene therapy which involves the manipulation of genes in cells that will be helpful to the patient but not inherited to the next generation. The other is germline gene therapy which involves the genetic modification of germ cells or the origin cells that will pass the change to the next generation. This lesson discussed the role of genes in heredity particularly how genes cause certain aspects in the growth and development of the individual. Parents especially look into the genetic traits that their children will exhibit (Prieto, 2019).

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